



UnionPay Integrated Circuit Card Specifications

— Product Specifications

Part I Contactless Reader Interface Specification



Version 2014

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1 Application Scope

This book applies to all UPI participants.

2 Normative References

The clauses in the following documents shall become the clauses of this standard after being quoted by this standard. For dated references, any subsequent amendment (excluding corrigenda) or revision is not applicable to this standard. However, parties coming to terms based on this standard are encouraged to investigate the possibility of applying the latest edition of these documents. For undated references, the latest edition of which is applicable to this standard.

ISO/IEC 7816-1:2011 *Identification cards -- Integrated circuit cards -- Part 1: Cards with contacts -- Physical characteristics*

ISO/IEC 7816-2:2007 *Identification cards -- Integrated circuit cards -- Part 2: Cards with contacts -- Dimensions and location of the contacts*

ISO/IEC 7816-3:2006 *Identification cards -- Integrated circuit cards -- Part 3: Cards with contacts -- Electrical interface and transmission protocols*

ISO/IEC 7816-4:2013 *Identification cards -- Integrated circuit cards -- Part 4: Organization, security and commands for interchange*

ISO/IEC 7816-5:2004 *Identification cards -- Integrated circuit cards -- Part 5: Registration of application providers*

Financial IC Card Pilot PSAM Card Application Specification

ISO 13491-1:1998 *Banking -- Secure cryptographic devices (retail) -- Part 1: Concepts, requirements and evaluation methods*

ISO 13491-2:2005 *Banking -- Secure cryptographic devices (retail) -- Part 2: Security compliance checklists for devices used in financial transactions*

ISO 7811-1:1995 *Identification cards -- Recording technique -- Part 1: Embossing*

ISO/IEC 7811-3:1995 *Identification cards -- Recording technique -- Part 3: Location of embossed characters on ID-1 cards*

ISO/IEC 7812-2:2000 *Identification cards -- Identification of issuers -- Part 2: Application and registration procedures*

ISO/IEC 14443-1:2000 *Identification cards -- Contactless integrated circuit(s) cards -- Proximity cards -- Part 1: Physical characteristics*

ISO/IEC 14443-2:2001 *Identification cards -- Contactless integrated circuit(s) cards -- Proximity cards -- Part 2: Radio frequency power and signal interface*

ISO/IEC 14443-3:2001 *Identification cards -- Contactless integrated circuit(s) cards -- Proximity cards -- Part 3: Initialization and anticollision*

ISO/IEC 14443-4:2001 *Identification cards -- Contactless integrated
circuit(s) cards -- Proximity cards -- Part 4: Transmission protocol*

3 Terms and Definitions

The following terms and definitions are applied to this Specification.

3.1 Contactless

Successfully exchange signals with card and provide energy supply for card without use of current conducting elements (in other words, no direct path from external interface device to the integrated circuit included in card).

3.2 Contactless Integrated Circuit(s) Card

It is an ID-1 type card (as specified in ISO/IEC 7810), in which the integrated circuit is installed and the communication with integrated circuit is completed in contactless way.

3.3 Contactless Card Reader

In the Specification, the readers are classified into two types: PCRs and non-PCRs. The PCRs refer to the contactless IC card reading-writing device that the reader can realize partial or all business logic of contactless IC card through secondary program development. The non-PCRs refer that the reader itself does not have secondary program development capability and can only receive the commands sent out by the equipment (such as POS, PC) connected to it, perform corresponding operations on card, indicator light, buzzer, LCD display etc. and return corresponding operation results.

3.4 Contactless Card

The contactless card defined in the Specification refers to the card compliant with the ISO14443 Specifications and loaded with UICS application.

3.5 Proximity Card (PICC)

It is an ID-1 type card, in which the integrated circuit and coupled circuit are installed and the communication with integrated circuit is completed via the inductive coupling with proximity coupling device.

3.6 Proximity Coupling Device (PCD)

It is the reading/writing device which supplies energy to PICC with inductive coupling and controls the data exchange with PICC.

3.7 IC Card Reader

It is the terminal equipment which can perform data exchange with IC card.

3.8 Terminal

It is the device which is compliant with UICS Specification and can realize the contactless card transaction operation with the card reader described in the Specification. It can be the equipment such as PC, POS, and refueling machine.

3.9 Collision

Two PICC transmissions in the same PCD exciting field and in the same time period disable PCD to identify which PICC sends out the data.

3.10 Anti-collision Loop

It is the algorithm utilized to prepare for the session between PCD and one or more among several PICCs in PCD exciting field.

3.11 Eject Proofing

When the card is suddenly pulled out or leaves the magnetic field in processing, the terminal shall remind the cardholder to insert or place the card again. After that, the terminal will check the issuer identification and the application sequence number to confirm whether the inserted card is the same one which was pulled out previously.

3.12 Block

It is the bytes sequence consisting of two or three fields defined as start field, information field and termination field.

3.13 PSAM

It is the terminal security access module, which is utilized to authenticate the validity of offline purchase transaction.

3.14 TYPE A

The TYPE A adopts an intermittent type modulation mode. In other words, when the information is indicated as “1”, the signal has been transmitted to the card; when the information is indicated as “0”, no signal is transmitted to the card. The interval is very short without any effect on normal operation of card. The advantages include visible information difference, few opportunities for interference, and not easy to make operational mistakes. The disadvantage is that a fluctuation may occur to the energy when it is required to continuously supply the energy to contactless card.

3.15 TYPE B

The TYPE B adopts a modulation mode with one amplitude modulation. In other words, the difference between information “1” and information “0” is that information “1” has a high signal amplitude, i.e. strong signal, while information “0” has a low signal amplitude, i.e. weak signal. Different information can be identified via the change in signal strength. The advantage is that the energy fluctuation will not occur during continuous signal transmission, while the disadvantages include unobvious information difference, relatively easy subjection to external interference and occurrence of error signal. It can also be remedied with through redundancy checks

4 Acronyms

Abbreviation	Interpretation
PCD	Proximity coupling device (reader)
PCRs	Programmable contactless card readers
non-PCRs	Non-programmable contactless card readers
PICC	Proximity card
ISO	International Organization for Standardization
AIP	Application Interchange Profile
DES	Data Encryption Standard
CA	Certification Authority
CAM	Online card authentication
CCCP	Common contactless communications protocol
CDA	Combined dynamic data authentication/application cryptogram generation
CVM	Cardholder Verification Method
DDA	Dynamic Data Authentication
DDOL	Dynamic Data Authentication Data Object List
IAC	Issuer Action Code
MTBF	Mean time between failures
qUICS	Quick UICS
RSA	Rivest, Shamir, Adleman
TAC	Terminal action code
TDEA	Triple data encryption algorithm; the DEA algorithm in this part refers to DES algorithm.

5 Descriptions of Message Symbols

The Specification describes the data type in the way of type (length).

The following data types are involved in the Specification:

A - Letters - 'A' to 'Z' and 'a' to 'z'. Its value is shown in the quotation marks, e.g. 'Sample'.

AN - Letters and numbers - 'A' to 'Z', 'a' to 'z' and '0' to '9'. This type of data is shown in the quotation marks, e.g. 'Sample'.

ANS - Letters, numbers and special characters - 'A' to 'Z', 'a' to 'z', '0' to '9' and special characters. This type of data is shown in the quotation marks, e.g. 'Sample'.

X - Hexadecimal system - One-byte unsigned hexadecimal system. Its value is from 0 to 255 and the prefix 0X is added before the value, e.g. 0X00.

6 Performance Requirements on Card Reader

6.1 Basic Performance Requirements on Card Reader

Table 1 Basic Performance Requirements on Card Reader

Radio-frequency compatibility standard	China Integrated Circuits (IC) Card Specification Part 11: Contactless IC Card Communication Specification
Radio-frequency operating frequency	13.56MHz±7KHz
Radio-frequency communication rate	106kbit/s
Serial interface baud rate	115200bps (default)
Communication interface	RS232. USB (optional)
Status Indication	LED indicator light (mandatory), buzzer (mandatory) and display (optional)
Working temperature	-5℃～50℃
Relative humidity	20%～95%
Contactless UnionPay logo area	Minimum 25mm (length) x 10mm (width); it can be enlarged at equal proportion and is posted in the visible position of front face.

6.2 Transaction Time

When performing the quick debit/credit (qUICS) payment transaction, the maximum time required for completing information exchange of card in the sensing area of card reader shall not exceed 500ms.

6.3 Magnetic Field Intensity Requirements

The contactless interface of card reader has the physical center as its center. As shown in Figure 1, the contactless interface is divided into two round areas with 1.5cm radius of inner circle and 2.5cm radius of outer circle. As shown in Figure 2, the cylindrical space is defined above the contactless interface. The magnetic field intensity requirement corresponding to individual points in this space is shown in Table 2.

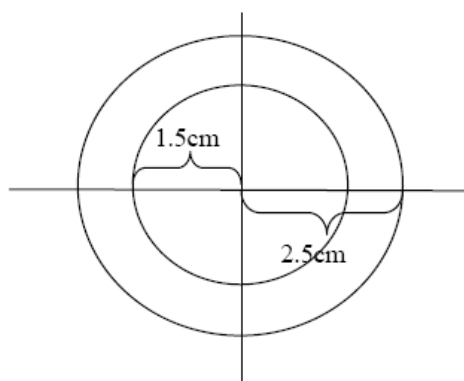


Figure 1 Planar division of contactless interface

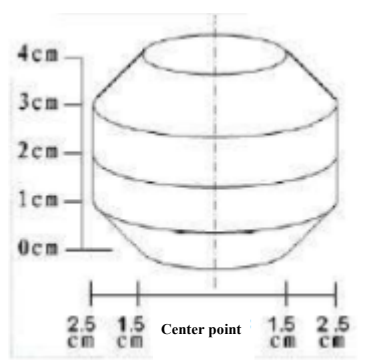


Figure 2 Spatial division of contactless interface

Table 2 Magnetic field intensity requirements

Height	Point 0	Inner circle	Outer circle	Field intensity requirements
Z=0cm	√	√	N/T	2.84A/m<HO<7.5 A/m
Z=1cm	√	N/T	√	2.79 A/m<HO<7.5 A/m
Z=2cm	√	N/T	√	2.74 A/m<HO<7.5 A/m
Z=3cm	√	N/T	√	2.54 A/m<HO<7.5 A/m
Z=4cm	√	√	N/T	2.33 A/m<HO<7.5 A/m

Note: The "√" indicates the magnetic field intensity requirement at the edge of this area and the "N/T" indicates that the magnetic field intensity of this area isn't defined.

6.4 Contactless Processing Chip

The contactless processing chip can process the TYPE A and Type B contactless chips complying with the stipulations in China Integrated Circuits (IC) Card Specification Part 11: Contactless IC Card Communication Specification. This protocol must remain compliant when complying with other communication protocols.

6.5 Display (Optional)

The card reader shall have a display to show the contents including transaction amount, transaction status, merchant discretionary information, etc., to the cardholder. The LCD display can have 2 lines or 4 lines with 32x122 lattice or 64x122 lattice.

According to the command sent out by financial terminal, the display should be able to show ASCII visible characters.

6.6 Status Indicator Light (Mandatory) and Buzzer (Mandatory)

The contactless card reader must inform the cardholder of transaction status in a simple and specific way via status indicator light, buzzer etc.

The status indicator light must be clearly visible to the cardholder in front of contactless card reader and the cashier behind contactless card reader. The indicator light has different representation modes depending on the operating status. The indicator light can also provide corresponding indication according to the command.

The buzzer shall be set with different sounds for different transaction status and shall be easily understood by the cardholder and the cashier.

The card reader includes the below 4 indicator lights:

No.	1	2	3	4
Color	Blue	Yellow	Green	Red

The correspondence among the indicator light, screen prompt information and buzzer and the transaction status is shown in the table below. Wherein, the indicator light status and buzzer status are mandatory requirements. The prompt information in the table below is for reference:

Table 3 Prompt information of card reader

Status	Meanings	Indicator light status	Prompt information	Beep
Not ready	The card reader has been powered on but has	All indicator lights are off.	Not ready, please wait.	No

Status	Meanings	Indicator light status	Prompt information	Beep
	not been connected to terminal, or the card reader has been connected to terminal and is performing two-way authentication with terminal.			
Idle status	The card reader has performed two-way authentication successfully, and the transaction can be performed.	The blue indicator light flashes once per 5 seconds, each period of lighting being around 200 milliseconds in duration, with other lights off.	UnionPay IC card	No
			Welcome your usage.	
Activating card	It is the process that terminal sends out the transaction amount to card reader after the cashier inputs the transaction amount on the terminal.	The blue indicator light stays on.	UnionPay IC card	No
			Purchase amount: Please swipe card.	
Transaction processing	(1) The card reader is reading the payment card data; or (2) the card reader has completed reading the payment card data and is reading non-payment application data.	(1) When the card reader only processes payment application, the blue indicator light remains on, while the yellow light also turns on; (2) when the card reader has completed reading the card payment data and is processing non-payment application data,	UnionPay IC card	No
			Purchase amount: Transaction is in process...	

Status	Meanings	Indicator light status	Prompt information	Beep
		the yellow indicator light must be on.		
Remove card	All data have been read (for offline transaction, the DDA verification is required after the card is moved out from sensing area).	The blue and yellow lights remain on, then green indicator light turns on so the three lights are on for at least 750 milliseconds before all three are turned off.	UnionPay IC card	Yes
			Please remove card and wait a moment	
Transaction successful	The offline DDA verification is successful, or the online authentication is successful.	Maintain the same state as card removal, and indicate successful transaction.	UnionPay IC card	Yes
		For online transaction, the green indicator light flashes until issuer authentication is completed.	Purchase amount:	
			Purchase balance:	
			Transaction is successful!	
Transaction failed	An error occurs in the transaction process.	The red indicator light remains on, and displays relevant error information, including: multiple card conflict, try contact type or magnetic stripe type transaction, or card not removed, etc.	UnionPay IC card	Yes
			Transaction failed! Error information [Error code]. Welcome your usage (Note: Wherein, the error prompt information can be shown according to the error type, e.g. 'Insufficient card balance', 'Please swipe	

Status	Meanings	Indicator light status	Prompt information	Beep
			card again'. If any error code is present, corresponding error code is required to be shown.)	

6.7 PIN Input Device (Optional)

If the IC card reader has the function of inputting PIN, the PIN input device on it must comply with the requirements specified in UnionPay Card Acceptance Terminal Security Specification Part VI: Security Specification for PIN Input Device of UnionPay Card Acceptance Terminal.

6.8 Supported Language

The card reader shall at least support the Chinese as its default language. The card reader supports the selection of its default language via terminal. The card reader can also support the simultaneous display of two languages.

6.9 CVM (Optional)

The card reader shall support the following cardholder verification methods (CVM) in a selective way.

- Signature
- Online PIN
- No CVMs

6.10 Reader Software

The reader shall have the device initialization, hardware self-checking and alarm functions. The reader shall have the capability to accept two types of cards including TYPEA and TYPEB cards.

6.11 Protocol Compatibility

When the reader communicates with the card, it shall comply with the stipulations in China Integrated Circuits (IC) Card Specification Part 11: Contactless IC Card Communication Specification.

If it is necessary to add other non-mandatory communication protocols to the reader, the processing of this communication protocol shall not be affected.

6.12 Polling Processing

The reader must be able to detect cards with different protocols via the polling mechanism. In other words, in order to detect the presence of contactless card entering effective action area of reader, the reader shall send out request signal repeatedly and determine whether there is card response. The IC card reader will alternately send out the request signal for Type A card and Type B card, and the commands and responses of Type A card and Type B card shall have no mutual interference.

For alternate mechanisms for sending out the request signal for Type A card and Type B card, no mandatory requirements are specified.

7 Card Reader and Terminal Protocol

This chapter defines the software protocol and command message format to be used for the communication between card reader and terminal. Card reader can be connected to the terminal in a variety of ways, including but not limited to serial port/USB/Bluetooth/ audio connection. This section describes in details the implementation of serial communication protocol used between card reader and terminal. Please refer to the related industry standard to get information about the communication standard for USB, bluetooth and audio communication. All business logic is realized in the contactless card reader. When it is necessary to set the reader or read/write the card, the terminal sends out corresponding commands to the reader, and the reader returns operation results.

If the terminal doesn't initiate payment transaction command, the card reader shall suspend card-seeking function temporarily.

Besides receiving card information from the card reader, terminal may be required to perform other functions. Therefore, it is essential to ensure that the message process won't significantly increase the transaction time.

7.1 Serial Interface

The card reader and the terminal shall support RS232. If necessary, the reader can also support USB communication mode. The operation schematic diagram of reader, card and terminal device is shown in the figure below:

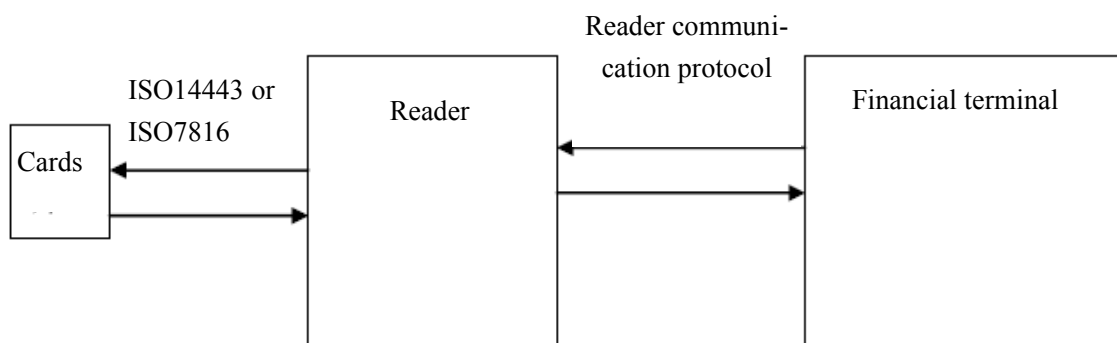


Figure 3 Operating mode of reader

The device supplier must provide the serial interface line sequence.

The example of serial interface line sequence diagram is shown in the figure below:

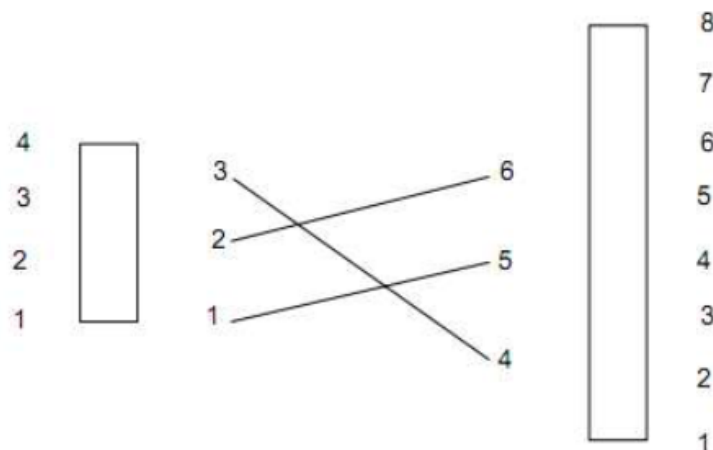


Figure 4 Serial interface line sequence diagram of reader

Table 4 Function of reader serial interface pin

Pin	Color	Function
1, 5	Red	TxD
2, 6	Black	RxD
3, 4	brown	Ground

7.2 Serial Interface Protocol

The terminal is connected to the card reader via RS232. As a transport protocol, the serial interface protocol takes the terminal and card reader as two points to be connected together. This serial interface protocol adopts 8 data bits, 1 start bit and 1 stop bit, and it doesn't adopt the data frame format with parity check bit. The communication rate can be set to one of the following rates:

- 115200 bps
- 57600 bps
- 38400 bps
- 19200 bps

The 115200bps communication rate is adopted as default. It is recommended to use the communication rate above 38400bps to achieve a better performance.

7.3 Protocol Descriptions

Ensure that the terminal and card reader are powered on and kept in mutually connected status before communication. The terminal determines the data connection via POLL command. The card reader can make the following responses:

- For response within the timeout range, confirm POLL_P.
- For two-way authentication response, execute POLL_A.

- For response rejection, execute POLL_N.
- If the card reader doesn't work, no response will be made.

The Poll and its response information (POLL_P, POLL_A and POLL_N) are defined in Chapter 10.

The two-way authentication is defined in Chapter 9.

The discussions on communication protocol are made in the several cases below:

7.3.1 Case 1 - Successful Communication

After the terminal is powered on, it sends POLL command to the card reader to establish data connection. If the card reader responds POLL_P within the specified period, the terminal will start to send the data including sequence number generated by the terminal to the card reader.

This process is shown in the figure below:

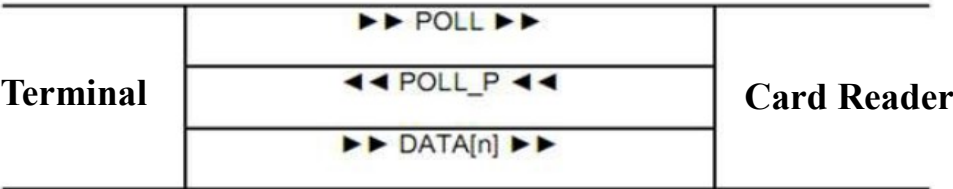


Figure 5 Successful communication process

The card reader will respond a response message with a sequence number of n+1, if it did not detect any data errors (e.g. CRC error, character frame format error).

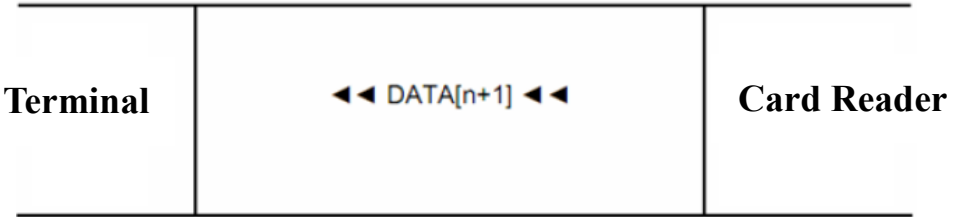


Figure 6 Data error handling process

If any error is detected, the card reader will respond the same data to the terminal and use the same sequence number. The terminal must check the transported and received sequence numbers after each request/response interaction. If the request/response interaction is correct, the terminal will perform the next communication with the sequence number of this request message plus 2.

7.3.2 Case 2 - Idle Time Sequence

The terminal must send POLL command to the card reader at the interval specified in second by the parameter P_POLL_MSG to detect whether the card reader is normally connected to the terminal. After the card reader receives POLL command, it shall respond POLL_P or POLL_A within the period specified in second by the parameter P_MSG_TIMEOUT.

The P_POLL_MSG and P_MSG_TIMEOUT are timeout parameters which shall be defined in Appendix A. The default parameters are 10s and 0.5s respectively.

After the terminal sends data block Data[n], it won't send POLL command any more, and it will wait for the card reader to respond data block Data[n+1] within the specified period.

After the contactless payment transaction is completed or the terminal cancels the payment transaction, the terminal shall send out the POLL command again.

Notice: The sequence number of POLL request and POLL response command is set to 0x00.

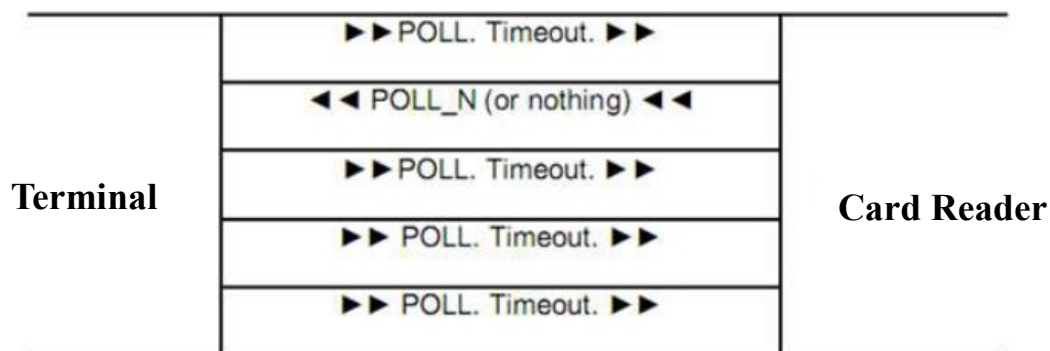
7.3.3 Case 3 - No Response from Card Reader

This case results from the following causes:

- The card reader isn't powered on
- The card reader is not operational
- The serial interface line isn't connected

The terminal sends POLL command to establish the data connection between the terminal and the card reader. If the card reader makes no response or respond POLL_N, the terminal will send POLL command again. If the card reader still makes no response after the POLL command is sent for the second time, the terminal will send POLL command to the card reader for the last time. If no effective response is received from the card reader, the terminal will automatically send POLL command at the interval specified in second by P_POLL_MSG to detect whether the card reader is present.

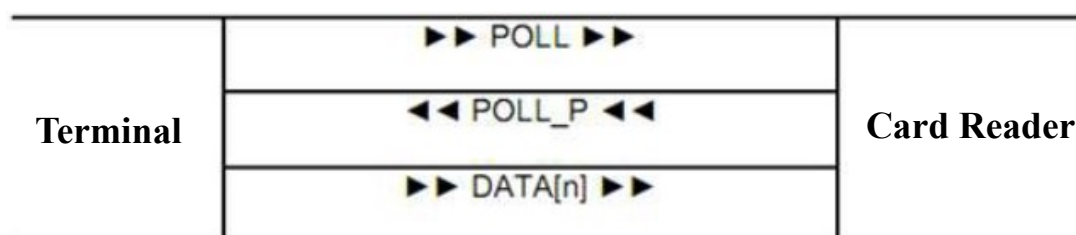
The relevant process is shown in the figure below:

**Figure 7 Process for no response of card****7.3.4 Case 4 - Card Reader Wake-up**

This case results from the following causes:

- The card reader power is suddenly interrupted and then powered on again.
- The serial interface connecting line is disconnected and then connected again.

As the terminal sends POLL command repeatedly at the interval specified in second by P_POLL_MSG to detect the presence of card reader, it can still detect the POLL_P response of card reader utilized to reestablish connection. The terminal will reestablish the data connection as per the process in Case 1.

**Figure 8 Card reader wake-up process****7.3.5 Case 5 - Terminal is Unready**

This case results from the following causes:

- The terminal isn't powered on
- The terminal isn't operational
- The terminal isn't connected to the card reader

As the card reader doesn't receive the POLL command utilized to establish data connection, the card reader shall stop card-seeking process and it must wait for the terminal to reestablish data connection so that normal operation can be resumed.

7.3.6 Case 6 - Data Block Error and Resending for Timeout

If the card reader detects any error in transportation process (such as wrong CRC check and wrong data frame format) it will return the terminal with the same data and sequence number as those of the terminal. Relevant procedure is shown below:

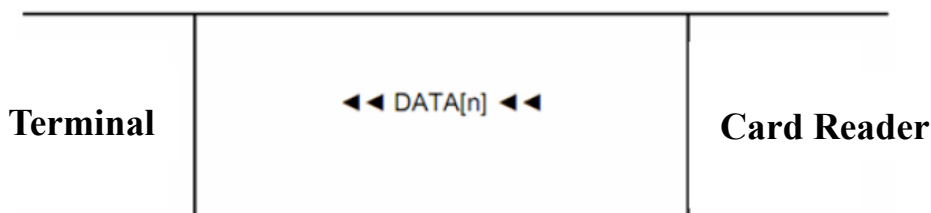


Figure 9 Process for data block error and timeout resending

After receiving the data block with the sequence number (n) from the card reader, the terminal must resend the corrected data block with the original sequence number n. If the card reader still detects any error in the data block, it will respond to the terminal with the same data and sequence number as those from the terminal. If no data block with the sequence number (n+1) is received from the card reader for successive three times, the terminal will stop sending request message and start to resend POLL command at the interval specified in second by P_POLL_MSG.

7.4 Data Block Format

All the messages transported between the terminal and the card reader shall adhere to the following format:

Fixed data head	Variable data field	Fixed data end
-----------------	---------------------	----------------

The composition of data head is shown in the table below:

STX	Serial No.	Command code	Data Length
-----	------------	--------------	-------------

When the data length is 0x00, it indicates that no data is included in the data field.

The composition of data end is shown in the table below:

CRC	ETX
-----	-----

The data block of request message is shown as follow:

Variable data

The response data block:

Response	Encrypted variable data
----------	-------------------------

code	
------	--

All request messages transported to the contactless card reader by the terminal shall adhere to the following format:

STX	Serial No.	Command code	Data Length	Variable data	CRC	ETX
-----	------------	--------------	-------------	---------------	-----	-----

All response messages transported to the terminal by the contactless card reader shall adhere to the following format. It shall be noticed that the highlighted variable data is encrypted.

STX	Serial No.	Command code	Data Length	Variable data	CRC	ETX
-----	------------	--------------	-------------	---------------	-----	-----

These fields will be described in Section 7.5 in detail.

7.5 Field Descriptions

7.5.1 STX

STX is the fixed value 0x02 in one byte and is the starting mark in the beginning of data block.

7.5.2 Sequence Number

The terminal generates a sequence number in the length of one byte with the range of 0x01 - 0xFF. When sequence number is 0xFF, the next sequence number to be generated by the terminal shall be reset to 0x01. This sequence number is only used in the authentication message and transaction message. If the response request sent by the terminal is successfully responded by the card reader, the next sequence number to be sent by the terminal shall be increased by two. 0x00 will be used as the sequence number for other types of message (as POLL command).

Notice: If the terminal sends the data block with a sequence number 0xFF to the card reader, the card reader can return the authentication response message or transaction response message with the sequence number 0x00.

7.5.3 Data Length

The data length has a size of two bytes and it is utilized to indicate and designate the variable data field length of request message and the response code and variable data length of response message. The sending and receiving buffers of terminal and card reader support at least 1024 bytes.

SET PARAMETERS is the only command with the possible data block length exceeding 1024 bytes at present. The terminal must prohibit the length of data block exceeding 1024 bytes. If the length of parameter to be updated exceeds 1024 bytes, the terminal must select to send several separate SETPARAMETERS commands.

7.5.4 Command Code

The command code indicates the command type of message and its size is one byte.

Individual command codes are shown as follows. See Chapter 8 for specific definitions.

Command type	Command code
POLL message	
POLL	0x07
Echo	0x08
Debugging and optimization message	
RFU Set debugging and optimizing message mode	0x11 0x10
Set Parameters	0x12
Authentication message	
Communication initialization	0x20
Two-way authentication	0x21
Key generation	0x22
Card reader shielding	0x23
Transaction message	
Quick debit/credit contactless transaction	0x30
Reset transaction	0x31
Display status	0x32
Transaction online post-processing	A
RFU	0x33 ~ 0x3F
Management message	
Entering management mode	0x40

Command type	Command code
Obtain properties	0x41
Set properties	0x42
Obtain time and date	0x43
Set time and date	0x44
Obtain parameters	0x45
Obtain serial interface communication rate	0x52
Set serial interface communication rate	0x53
Reset acquirer key	0x54
Resume card reader	0x55
Obtain UICS tag value	0x56
Set UICS tag value	0x57
Obtain display information	0x58
Set display information	0x59
Obtain CVM properties	0x5A
Set CVM properties	0x5B
Set UICS public key	0x61
Search for UICS public key	0x62
Set revocation public key certificate	0x63
Search for revocation public key certificate	0x64
Set blacklist	0x65
Search blacklist	0x66
Set UICS fixed parameters	0x67
Set AID related parameters	0x68
RFU	0x70 ~ 0x7F

7.5.5 Response Code

With a size of one byte, it is returned in the response message for specific request and indicates the response information.

Individual errors and response codes are specified in Appendix B.

7.5.6 Variable Data Field

For data field, message of different type adopts different length.

The TDEA encryption is performed on the data in response information in ECB mode. If the byte size of data field isn't integral multiples of 8, 0x00 will be added after the data field to complement its byte size to integral multiples of 8.

7.5.7 Cyclic Redundancy Check

The standard CRC-16 check is adopted, and the polynomial is shown below:

$$G(X) = X^{16} + X^{15} + X^2 + 1$$

The calculation range of CRC is from the sequence number to the variable data field.

7.5.8 ETX

ETX occupies a length of one byte and its fixed value is 0x03. It is used as the identification for end of data block.

7.6 Verification of Request Message

When the card reader receives the request information sent by the terminal, the following checks will be performed on the information:

- Whether the length of message is correct?
- Whether the calculated CRC is consistent with the CRC of data block?

7.7 Verification of Response Message

When the terminal receives the response sent by the card reader, the following checks will be performed on the information:

- Whether the length of message is correct?
- Whether the sequence number of message is correct?
- Whether the calculated CRC is consistent with the CRC of data block?

8 Message/Command Type

The terminal can start and initiate the following command messages:

- POLL message
- Echo message
- Debugging and optimization message
- Authentication message
- Transaction message
- Management message
- All command messages shall be described in detail as follows. The card reader shall be designed to be compatible with other command messages which may be added in future.

8.1 POLL Message

The terminal sends POLL message to the card reader to establish data connection and judge the presence of card reader.

8.2 Echo Message

This command is used to help the terminal and card reader manufacturers to detect whether the terminal and card reader have established correct data connection. If any data is included in the data field of the Echo message sent by the terminal, the card reader must respond the same data to the terminal.

8.3 Debugging and Optimization Message

Before the terminal sends debugging and optimization message, the PIN must be verified at the terminal.

If the card reader is in the debugging and optimization mode, it can receive SET PARMENTERS command;

Unless any failure treatment is required, the card reader can't enter the optimization mode in the production environment.

8.3.1 Set Debugging and Optimization Mode Command

This command is used to make the card reader enter debugging and optimization mode or return to normal mode.

8.3.2 Set parameter Command

The terminal can change the parameters of card reader with set parameter command, and optimize the properties of card reader and terminal with parameter configuration. This command must be able to be run only after the debugging and optimization mode set command is successfully executed.

8.4 Authentication Message

8.4.1 Communication Initialization Command

This command is used to initiate secure communication and allow the card reader to generate the authentication session key used for two-way authentication.

8.4.2 Two-way Authentication Command

This command is used for mutual identification and verification between the terminal and the card reader. It is used to confirm the authenticity of the terminal and card reader. The contactless payment transaction is allowed only after two-way authentication. And the management message is allowed to be executed only after the two-way authentication.

8.4.3 Key Generation Command

This command is used to generate acquirer working key and session key.

Relevant operation method for key is discussed in Chapter 9.

8.4.4 Card Reader Shielding Command

If the terminal detects that the card reader is invalid (for example, the identification of card reader is wrong), the terminal will send the card reader shielding command to this card reader. After receiving the shielding command, the card reader shall prohibit card-seeking function, clear buffer zone and erase all stored acquirer keys.

8.5 Transaction Message

8.5.1 Quick Debit/Credit Contactless Transaction Command

The terminal sends this command to the card reader to instruct the card reader that the terminal is ready to receive PICC data and perform quick debit/credit contactless transaction.

If any PICC enters the sensing area of card reader, the card reader shall read the card and return the data of card to the terminal. The card reader can run card-seeking function only after receiving this command.

8.5.2 Complete UICS Debit/Credit Transaction (Optional)

The terminal sends this command to the card reader to instruct the card reader that the terminal has been prepared to receive PICC data and perform the complete process of UICS debit/credit transaction. If any PICC enters the sensing area of card reader, the card reader shall read the card, return the data of card to the terminal and select corresponding application.

The card reader will also run card-seeking function after receiving this command.

8.5.3 Display Status Command

The terminal uses this command to inform the card reader to show specific status to the user. Two different statuses are defined as follows:

- **Successful** The transaction has been approved.
- **Failed** The transaction isn't approved, with all error responses.

The terminal and card reader shall display relevant prompt information used to prompt the cardholder. Different prompt information can be defined for different merchants. Refer to Section 14.13 for relevant details to obtain the section of information displaying command.

8.5.4 UPCARD Transaction Command

The terminal sends this command to the card reader to instruct the card reader that the terminal is ready to receive the data of UPCARD. If any UPCARD enters the sensing area of card reader, the card reader shall read the card and return the data of card to the terminal. The card-seeking function can be run only after the card reader receives this command.

8.6 Management Message

When a management message is sent, it is essential to verify the PIN at the terminal first.

In the production environment, if it is required to run these commands, the two-way authentication of terminal and card reader must be performed.

Individual commands for management message are shown in the table below:

Table 5 Management message commands

Management message commands	Command code	Description
Entering management mode	0x40	It is used to make the card reader enter management mode and to obtain the factory information, firmware and version number of card reader. (The successful execution of this command is the precondition for executing all the following management message commands)
Obtain properties	0x41	It is used to search for the payment scheme and sub-scheme which can be supported by the card reader
Set properties	0x42	It is used to activate or release one payment scheme or sub-scheme which can be supported by the card reader

Management message commands	Command code	Description
Obtain time and date	0x43	It is used to obtain the time and date of card reader
Set time and date	0x44	It is used to set the time and date of card reader
Obtain parameters	0x45	It is used to obtain a series of predefined parameter values of card reader
Obtain serial interface communication rate	0x52	It is used to search for the serial interface communication rate set in the card reader
Set serial interface communication rate	0x53	It is used to set the serial interface communication rate of card reader
Reset acquirer key	0x54	It is used to clear the acquirer key in the card reader.
Resume card reader	0x55	Re-activate the card reader
Obtain UICS tag value	0x56	It is used to obtain the UICS data element tag data supported by the card reader.
Set UICS tag value	0x57	It is used to set the UICS data element tag data supported by the card reader.
Obtain display information	0x58	It is used to obtain the prompt information stored in the card reader
Set display information	0x59	It is used to set the prompt information stored in the card reader
Obtain CVM properties	0x5A	It is used to obtain the CVM properties of card reader
Set CVM properties	0x5B	It is used to activate or prohibit the CVM properties of card reader
Set UICS public key	0x61	It is used to download the UICS CA public key to the card reader or to clear the public key
Search for UICS public key	0x62	It is used to obtain the UICS CA public key loaded in the card reader
Set revocation public key certificate	0x63	It is used to revoke the public key certificate in the card reader or to clear it
Search for revocation	0x64	It is used to obtain the revocation public key

Management message commands	Command code	Description
public key certificate		certificate loaded in the card reader
Set blacklist	0x65	It is used to download the blacklist to the card reader or to clear the blacklist
Search blacklist	0x66	It is used to obtain the blacklist loaded in the card reader
Set UICS fixed parameters	0x67	It is used to set the fixed parameters related to UICS transaction
Set AID related parameters	0x68	It is used to set the transaction parameters related to AID

9 Encryption Service

This chapter describes the algorithm for two-way authentication between terminal and card reader, the generation method of key for authentication, and the method to encrypt data.

There are two types of two-way authentication. The terminal and the card reader must realize these two types of authentication, so as to:

Realize contactless payment transactions;

Execute management message relevant commands.

These two types of two-way authentication are protected by two types of key. For encryption, the symmetric double-length data cryptographic technique shall be applied. The card reader supplier preloads the manufacturer default key for each acquirer. This default key is used as the initial key of card reader.

When an acquirer receives the card reader, the acquirer must use its own key to replace the manufacture key of card reader. This chapter describes relevant details of encryption service. In addition, the following goals can be realized with these details:

- All the messages transported between the terminal and the card reader are not tampered with and can be verified.
- Messages transported between the terminal and the card reader can't be duplicated.

For the convenience of reading, the key acronyms involved in this chapter are listed as follows:

Key Acronyms	Meanings
IMEK _{MDK}	Initial message encryption key (manufacturer default key)
IAEK _{MDK}	Initial access encryption key (manufacturer default key)
IMEK	Initial message encryption key
IAEK	Initial access encryption key
MEK	Message encryption key
Msession	Message encryption session key
AEK	Access encryption key
Asession	Access encryption session key

9.1 Manufacturer Default Key

Before the sale of card reader, the card reader supplier shall preload the following two types of double-length key into the card reader:

- Initial message encryption key (manufacturer default key) IMEK_{MDK}
- Initial access encryption key (manufacturer default key) IAEK_{MDK}

When an acquirer receives the card reader, the acquirer must use its own key to replace the manufacture key of card reader according to the requirements in the below section.

9.2 Acquirer Key

All the terminals and card readers must be able to store at least the following double-length acquirer keys in a secure storage environment (as SAM card):

- Initial message encryption key (IMEK)
- IMEK authentication session key
- Message encryption key (MEK)
- MEK authentication session key
- Message encryption session key (Msession)
- Initial access encryption key (IAEK)
- IAEK authentication session key
- Access encryption key (AEK)
- AEK authentication session key
- Access encryption session key (Asession)

9.3 IMEK and IAEK

The acquirer must replace IMEK_{MDK} with IMEK. For individual acquirers, IMEK is different and it must be loaded into the terminal in a secure environment. The acquirer can use several groups of IMEK, so that the whole system won't be damaged if one IMEK is attacked.

The IMEK_{MDK} authentication session key is used for the two-way authentication in the first connection between the terminal and the card reader. In the follow-on session, IMEK_{MDK} will be replaced by IMEK.

IAEK_{MDK} will be replaced by IAEK in the same way.

Use the key generation command to replace IMEK_{MDK} and IAEK_{MDK} with IMEK and IAEK respectively. IMEK is encrypted by IMEK_{MDK} for transmission. IAEK is encrypted by IAEK_{MDK} for transmission.

9.4 MEK and MEK Session Key

IMEK is the master key of MEK. As mentioned before, the key generation command is used to replace IMEK_{MDK} and IAEK_{MDK} with IMEK and IAEK respectively.

In addition, the key generation command is also used to generate MEK from IMEK.

9.4.1 First Power-on Connection

For the first power-on connection between the terminal and the card reader, after the two-way authentication between terminal and card reader is performed, the IMEK_{MDK} authentication session key will be generated and the IMEK_{MDK} and IAEK_{MDK} will be replaced by IMEK and IAEK respectively. In addition, the terminal must generate the MEK and Msession key.

The terminal adopts the key generation command to generate MEK, encrypts and transmits MEK with the IMEK_{MDK} authentication session key. Then, the key generation command is used to generate Msession and the IMEK_{MDK} authentication session key is used to encrypt and transmit Msession.

Notice: If the IMEK_{MDK} authentication session key is lost, the IMEK authentication session key will be generated with the two-way authentication of IMEK between the terminal and the card reader. This key can also be used for the encryption and transmission of MEK.

9.4.2 Follow-on Power-on Connection

In the follow-on power-on connection process, the MEK authentication session key will be generated, which can be used to ensure the secure transmission of Msession between the terminal and the card reader. The contactless payment solution can be realized only when the Msession key is generated and the Msession key can be securely shared between the terminal and the card reader.

The terminal and the card reader must clear all session keys (as MEK authentication session key and Msession) when de-energized.

9.5 AEK and AEK Session Key

9.5.1 First Power-on Connection

After the first power-on connection between the terminal and the card reader, the two-way authentication between the terminal and the card reader shall generate the IAEK_{MDK} authentication session key. The terminal adopts the key generation command to generate AEK, encrypts and transmits AEK with the IAEK_{MDK} authentication session key. Then, the key generation command is used to generate Asession, and the IAEK_{MDK} authentication session key is used to encrypt and transmit Asession.

Notice: If the $IAEK_{MDK}$ authentication session key is lost, the two-way authentication of $IAEK$ can be performed between the terminal and the card reader to generate the $IAEK$ authentication session key. This key can also be used for the encryption and transmission of AEK .

9.5.2 Follow-on Power-on Connection

In the follow-on power-on connection process, the AEK authentication session key will be generated. This key can be used to ensure the secure transmission of $Asession$ between the terminal and the card reader.

The execution of management mode command requires a secure access environment. With the protection of AEK authentication session key, the data related to the command is encrypted by the $Asession$ key.

The management mode command can be executed in the production environment only when the $Asession$ key of terminal and card reader is successfully generated and $Asession$ is securely shared.

9.6 Key Type

Two parameters are used in Table 6. These two parameters are the identification of acquirer key in message.

Table 6 Key type

Key	Key type (Fixed)	Index number of key (variable)	Key usage
$IMEK_{MDK}$	00	00	- It is preloaded at factory by the card reader manufacturer - Generate the $IMEK_{MDK}$ authentication session key for authentication
$IMEK$	01	00	- Overlay $IMEK_{MDK}$ - Generate the $IMEK$ authentication session key for authentication - Generate MEK
MEK	02	01	- Generate the MEK authentication session key for authentication - Generate the $Msession$ for data encryption
$Msession$	03	01	- Data Encryption - The contactless card transaction must be performed only after $Msession$ is successfully generated and authenticated

Key	Key type (Fixed)	Index number of key (variable)	Key usage
IAEK _{MDK}	04	00	- It is preloaded at factory by the card reader manufacturer - Generate the IAEK_{MDK} authentication session key for authentication
IAEK	05	00	- Overlay IAEK_{MDK} - Generate the IAEK authentication session key for authentication - Generate AEK - It is used only in the case of the first connection between terminal and card reader
AEK	06	01	- Generate the AEK authentication session key for authentication - Generate the Asession for data encryption
Asession	07	01	- Data Encryption - The management mode must be entered only after Asession is successfully generated and authenticated

The parameter key type is used to identify individual different keys. The index number of parameter key allows the acquirer to use several IMEK and IAEK keys based on the consideration on security.

The value of key type is fixed and the index number of key is variable. The specific stipulations are as follows:

For IMEK and IAEK, if the acquirer wants to use several IMEK and IAEK, it can use other values such as 1, 2, 3, 4. It is recommended that the acquirer can use a maximum of 5 IMEK.

9.7 Two-way Authentication and Key Generation Algorithm

The terminal and the card reader adopt IMEK to perform authentication in the following process.

9.7.1 Authentication Performed with IMEK

The terminal generates the 8-byte random number RND_B and sends it to the card reader in the form of plaintext.

The card reader also generates the 8-byte random number RND_R. The card reader uses the IMEK key and calculates the IMEK authentication session key with the parameters RND_B and RND_R:

IMEK authentication session key = TDEA ((IMEK, RND_R(5:8), RND_B(1:4), RND_R(1:4), RND_B(5:8)). Wherein, RND_R (5:8) represents the 5th - 8th bytes of RND_R.

Then, the card reader adopts the algorithm 1 and encrypts RND_B and RND_R with the IMEK authentication session key.

Encryption algorithm 1:

TDEA (IMEK authentication session key, RND_B, RND_R)

Then, the card reader sends the cryptogram result generated with algorithm 1 and the RND_R in the form of plaintext to the terminal. The terminal also uses the IMEK key and calculates the IMEK authentication session key with the parameters RND_B and RND_R. In addition, it decrypts the cryptogram sent by the card reader with the generated IMEK authentication session key. The terminal will compare RND_B and RND_R, if they are consistent with the RND_B and RND_R previously sent to the terminal by the card reader, the card reader authentication shall be considered as successful.

After the terminal successfully authenticates the card reader, it uses the encryption algorithm 2 to encrypt RND_B and RND_R and sends the cryptogram obtained to the card reader.

Encryption algorithm 2:

TDEA (IMEK authentication session key, RND_B(5:8), RND_R(1:4), RND_B(1:4), RND_R(5:8))

If the card reader received the cryptogram, it shall decrypt the received cryptogram and obtain the RND_B and RND_R. If the RND_B and RND_R are correct, the terminal will be acknowledged.

If the two-way authentication failed, the card reader shall return error information.

For IMEK_{MDK}, this two-way authentication algorithm is still applicable.

9.7.2 Generation of MEK

After the terminal and the card reader succeed in the two-way authentication with the IMEK authentication session key, the terminal will generate MEK.

The terminal will generate one random number RND_M which will be used as MEK. The terminal will use the IMEK authentication session key to encrypt RND_M.

Then, the encrypted data will be sent to the card reader. The card reader will use the IMEK authentication session key to decrypt the data sent from the terminal. The decrypted data is MEK, and the card reader will store this MEK in a secure area.

Notice:

In order to prevent the whole system from being influenced due to the disclosure of IMEK, IMEK can't be directly used for authentication and encryption.

With the same algorithm as the one used to generate MEK, replace $IMEK_{MDK}$ with IMEK. The difference is that the terminal does not need to generate the random number RND_M. The terminal replaces RND_M with IMEK.

9.7.3 Two-way Authentication with MEK

The terminal generates the random number RND_B and sends it to the card reader in the form of plaintext.

The terminal generates the 8-byte random number RND_B and sends it to the card reader in the form of plaintext.

The card reader also generates one 8-byte random number RND_R. The card reader uses the MEK key and calculates the MEK authentication session key with the parameters RND_B and RND_R:

MEK authentication session key = TDEA (MEK, RND_R (5:8), RND_B(1:4), RND_R(1:4), RND_B(5:8))

Wherein, the RND_R (5:8) represents the 5th - 8th bytes of RND_R.

Then the card reader uses the algorithm 1 and encrypts RND_B and RND_R with the MEK authentication session key.

Encryption algorithm 1:

TDEA (MEK authentication session key, RND_B, RND_R)

Then, the card reader sends the cryptogram result generated with algorithm 1 and the RND_R in the form of plaintext to the terminal. The terminal also uses the MEK key and calculates the MEK authentication session key with the parameters RND_B and RND_R. In addition, it uses the generated MEK authentication session key to decrypt the cryptogram sent from the card reader. The terminal will compare RND_B and RND_R, if they are consistent with the RND_B and RND_R previously sent to the terminal by the card reader, the card reader authentication will be considered as successful.

After the terminal successfully authenticates the card reader, it uses the encryption algorithm 2 to encrypt RND_B and RND_R and sends the cryptogram obtained to the card reader.

Encryption algorithm 2:

TDEA (MEK session key, RND_B (5:8), RND_R (1:4), RND_B(1:4), RND_R(5:8))

After the card reader received the cryptogram, it will decrypts it and obtain the RND_B and RND_R. If RND_B and RND_R are verified as correct, the terminal will be acknowledged.

The transaction message can be executed only after the terminal and the card reader complete the two-way authentication.

9.8 Generation of Session Key

In each power-on process, after the two-way authentication is executed and generated, new session key will be generated for the follow-on data encryption with the key generation command.

9.8.1 Generation of Msession

After MEK is generated, the Msession will be generated in each power-on process.

After the terminal and the card reader are mutually connected and the two-way authentication with MEK is successful, the terminal shall generate the random number RND_S used as Msession first, and then, after encrypting the RND_S with the MEK authentication session key, it will send it to the card reader. The card reader decrypts the data sent from the terminal, and then stores it in a secure area. The card reader must clear the Msession every time it is de-energized.

9.9 Generation of AEK and Asession

The generation of AEK and Asession adopts the same method as the one for generation of MEK and Msession.

9.10 Encryption of Data

Only the data of data field is encrypted. The encryption method adopts the standard ECB mode TDEA encryption. If the byte size of data field isn't integral multiples of 8, 0x00 will be used to complement the byte size of data field to integral multiples of 8. Msession is used to encrypt the whole variable-length data field in transaction message, and Asession is used to encrypt the whole variable-length data field in management message.

9.11 Generating New MEK and AEK

If it is suspected that MEK and AEK are lost or disclosed due to some reasons, new MEK and AEK can be generated. The terminal can clear the key of card reader by sending acquirer key reset command (RESET ACQIRER KEY).

As described in Section 9.7, the terminal and card reader will use IMEK/IAEK to perform two-way authentication and generate new MEK and AEK (See the descriptions in Section 9.7).

The RESET ACQUIRER KEY command must be executed in a secure access environment.

9.12 Security Measure for Key Management

1. Secure and Controllable Environment

Three types of messages defined in the Specification are:

- 1) Transaction message
- 2) Management message
- 3) Debugging and optimization message

These three types of messages can be run in a secure and controllable environment. The secure and controllable environment refers to the places which can't be accessed without authorization, such as the workplace of acquirer or the workshop building of terminal supplier.

The terminal manufacturer and the acquirer can perform routine test, set card reader parameter or debugging activation with the management message or debugging and optimization message. Before the acquirer adopts the management message or debugging and optimization message, the PIN set by the terminal manufacturer must be verified.

2. Merchant Operating Environment

In the merchant operating environment, only the transaction message can be used. After the terminal and the card reader are both powered on, they will perform two-way authentication by using the Msession key for improvement of transaction message security. When the management message is used, Asession is utilized to improve the security. However, no Asession is used in transaction.

The merchant isn't allowed to use the management message and debugging and optimization message. After Asession is used for two-way authentication, it is required to verify the PIN to trigger this type of message. If it is necessary to update the configuration of device, the acquirer must ensure that only the authorized person is allowed to access the device.

10 POLL, Echo and Optimization Messages

This chapter describes the structure of request packet and response packet of POLL, Echo and optimization messages.

The command codes of POLL, Echo and optimization messages are shown as follows:

Command type	Command code
POLL message	
POLL	0x07
Echo	0x08
Debugging and optimization message	
Set debugging and optimization message mode	0x10
RFU	0x11
Set Parameters	0x12

10.1 POLL

The terminal sends POLL message to the card reader to establish the data connection between the terminal and the card reader and judges whether the card reader has been connected to the card reader.

Table 7 POLL command message

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	0x00
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x07
Data Length	Length of request message	0x00 0x00
Data field	Contain no data	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

10.1.1 POLL - Response Code

Table 8 POLL command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	0x00
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x07
Data Length	Length of response message	0x00 0x00
Response code	Response code of card reader	The following situations are possible: • RC_POLL_P • RC_POLL_N • RC_POLL_A
Data field	Contain no data	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

10.2 Echo

As a debugging command, Echo is usually only used in the development and testing. The Echo command has no effect on the process of transaction message and management message.

This message is used by the suppliers for terminal and card reader to detect whether the data connection of terminal and card reader is established. If any data is contained in the data field sent by the terminal, the card reader will return the same data in the response message to the terminal.

Table 9 Echo command

Field	Meanings	Value
STX	Start of message	0x02

Field	Meanings	Value
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x08
Data Length	Length of request message	Variable
Data field	Test data	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 10 Echo command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x08
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE
Data field	Feedback the received data.	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

10.3 Set Debugging and Optimization Mode

This command is used to enable the debugging and optimization mode of card reader.

Table 11 Set debugging and optimization mode command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x10
Data Length	Length of request message	0x01
Data field	X(1) - 0x00 Enable debugging and optimization mode 0x01 Enable normal mode	X(1)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 12 Set debugging and optimization mode command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x10
Data Length	Length of response message	0x02
Response code	Response code of card reader	X(1) One of the following values: - RC_SUCCESS - RC_ACCESS_FAILURE - RC_INVALID_DATA - RC_ACCESS_NOT_PERFORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

10.3.1 Set Parameter

In the development and testing phase, the set parameter command is used to set the parameters of card reader. See Appendix A for the parameters preloaded at factory.

Table 13 Set parameter command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x12
Data Length	Length of request message	Variable
Data field	The format of parameter is as follow: - X(2) - 0x0000 - 0xFFFF number of parameters - Several groups of the following data are present depending on the number of parameters: - X(2) - Parameter Index Number - X(2) - Length of parameter - X(2) - Parameter	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 14 Set parameter command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)

Field	Meanings	Value
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x12
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	X(1) One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_DATA • RC_NO_PARAMETER
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

11 Authentication Message

This chapter describes the structure of request packet and response packet of communication initialization command, two-way authentication command, key generation command and card reader shielding command.

Authentication message	
Communication initialization	0x20
Two-way authentication	0x21
Key generation	0x22
Card reader shielding	0x23

The data in the data field of two-way authentication command and key generation command has been encrypted with proper key, so it isn't necessary to encrypt it additionally.

11.1 Communication Initialization

The communication initialization command is used to initialize the secure communication between the terminal and the card reader. The terminal is allowed to authenticate the card reader and generate the IMEK/IAEK authentication session key for mutual authentication.

Table 15 Communication initialization command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x20
Data Length	Length of request message	0x00 0x0A
Data field	- Key type: - 0x00 IMEK_{MDK} - 0x01 IMEK - 0x02 MEK - 0x04 IAEK_{MDK} - 0x05 IAEK - 0x06 AEK	X(0x0A)

Field	Meanings	Value
	- Index of key - Random number X(8) generated by the terminal	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 16 Communication initialization response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x20
Data Length	Length of response message	0x00 0x1B
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE - RC_INVALID_DATA - RC_AUTH_FAILURE
Data field	- Key type: - 0x00 IMEK_{MDK} - 0x01 IMEK - 0x02 MEK -0x04 IAEK_{MDK} - 0x05 IAEK - 0x06 AEK - Index of key - Random number X(8) generated by the card reader	X(1A)

Field	Meanings	Value
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

11.2 Two-way Authentication

The two-way authentication command allows:

- Terminal authentication of card reader
- Card reader authentication of terminal

Table 17 Two-way authentication command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x21
Data Length	Length of request message	0x10
Data field	• Key type: -0x00 IMEK_{MDK} -0x01 IMEK -0x02 MEK -0x04 IAEK_{MDK} -0x05 IAEK -0x06 AEK • Key index X(1) • Random numbers RND_B and RND_R X(10) encrypted with authentication session key according to algorithm 2	X(0x12)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 18 Two-way authentication response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x21
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE
		• RC_INVALID_DATA • RC_AUTH_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

11.3 Key Generation

The key generation command is used to:

- Replace $IMEK_{MDK}$ and $IAEK_{MDK}$ with $IMEK/AMEK$
- Generate MEK/AEK
- Generate $Msession$ and $Asession$.
- Replace the previous $IMEK/IAEK$ with new $IMEK/IAEK$, if necessary
- Replace the previous MEK/AEK with new MEK/AEK , if necessary

Notice:

The key type and key index used by this command are different from those of communication initialization command and two-way authentication command.

For key generation command, the key type and key index indicate which type of key will be generated; for the communication initialization command and two-way authentication command, the key type and key index indicate which type of key will be used for the authentication in communication.

Table 19 Key generation command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x22
Data Length	Length of request message	0x00 0x12
Data field	- Key type: -0x01 IMEK -0x02 MEK -0x03 Msession -0x05 IAEK -0x06 AEK -0x07 Asession - Key index X(1) - Encrypted random number X(10) Notice: The key type and key index determine which type of key is generated with the random number. If the key type is 0x00 or 0x04, this random number will be IMEK or IAEK. These values are predefined by the acquirer. The terminal or terminal management system must provide the acquirer with an interface for inputting IMEK and IAEK.	X(0x12)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 20 Key generation command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01

Field	Meanings	Value
Command code	Command type identifier	0x22
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_AUTH_NOT_PERFORMED • RC_INVALID_KEYINDEX
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

11.4 First Power-on Processing

11.4.1 Assume that the card reader has been loaded with $IMEK_{MDK}$ and $IAEK_{MDK}$

The following steps are applied to replace $IMEK_{MDK}$.

Step 1 Communication initialization command

Use $IMEK_{MDK}$, the key type is 0x00 and the key index is 0x00

=>Generate $IMEK_{MDK}$ authentication session key

Step 2 Two-way authentication command

Use $IMEK_{MDK}$, the key type is 0x00 and the key index is 0x00

=>Use $IMEK_{MDK}$ authentication session key for authentication

Step 3 Key generation command

Generate $IMEK$, the key type is 0x01 and the key index is 0x00

=>Use $IMEK_{MDK}$ authentication session key to transmit the predefined $IMEK$

Step 4 Key generation command

The terminal still adopts the key generation command to generate MEK , with the key type is 0x02 and the key index is 0x01

=> Use MEK transmitted by IMEK_{MDK} authentication session key

Step 5 Key generation command

The terminal still adopts the key generation command to generate Msession, with the key type is 0x03 and the key index is 0x01.

=>Use IMEK_{MDK} authentication session key to transmit Msession

Note: If the IMEK_{MDK} authentication session key is lost

Generate MEK and Msession as per the following steps:

Step 1 Communication initialization command

Use IMEK, the key type is 0x01 and the key index is 0x00

=>Generate IMEK authentication session key

Step 2 Two-way authentication command

Use IMEK, the key type is 0x01 and the key index is 0x00

=>Use IMEK authentication session key for authentication

Step 3 Key generation command

Generate MEK, the key type is 0x02 and the key index is 0x01

=>Use MEK transmitted by IMEK authentication session key

Step 4 Key generation command

Generate Msession. The key type is 0x03 and the index number of key is 0x01.

=>Use IMEK authentication session key to transmit Msession

Reset IAEK_{MDK} and generate AEK and Asession as per the following steps:

Step 1 Communication initialization command

Use IAEK_{MDK} , the key type is 0x04 and the key index is 0x00

=>Generate IAEK_{MDK} authentication session key

Step 2 Two-way authentication command

Use $IAEK_{MDK}$, the key type is 0x04 and the key index is 0x00

=>Use $IAEK_{MDK}$ authentication session key for authentication

Step 3 Key generation command

Generate $IAEK$, the key type is 0x05 and the key index is 0x00

=>Use $IAEK_{MDK}$ authentication session key to transmit the preloaded $IAEK$

Step 4 Key generation command

The terminal still adopts the key generation command to generate AEK , with the key type is 0x06 and the key index is 0x01

=>Use AEK transmitted by $IAEK_{MDK}$ authentication session key

Step 5 Key generation command

The terminal still adopts the key generation command to generate $A_{session}$, with the key type is 0x07 and the key index is 0x01

=>Use $IAEK_{MDK}$ authentication session key to transmit $A_{session}$

11.5 Follow-on Power-on Processing

Assume that the $IMEK$, $IAEK$, $MEK1$ and $AEK1$ have been generated and can be shared between the terminal and the card reader.

11.5.1 Generating $M_{session}$

Step 1 Communication initialization command

Use $MEK1$, with the key type is 0x02 and the key index is 0x01

=>Generate $MEK1$ authentication session key

Step 2 Two-way authentication command

Use $MEK1$, with the key type is 0x02 and the key index is 0x01

=>Use $MEK1$ authentication session key for authentication

Step 3 Key generation command

Generate Msession. The key type is 0x03 and the index number of key is 0x01.

=>Use MEK1 authentication session key to transmit Msession

11.5.2 Generating Asession

Step 1 Communication initialization command

Use AEK1, with the key type is 0x06 and the key index is 0x01

=>Generate AEK1 authentication session key

Step 2 Two-way authentication command

Use AEK1, with the key type is 0x06 and the key index is 0x01

=>Use AEK1 authentication session key for authentication

Step 3 Key generation command

Generate Asession, with the key type is 0x07 and the key index is 0x01

=>Use AEK1 authentication session key to transmit Asession

11.5.3 Replacing IMEK

If necessary, the acquirer can replace the previous IMEK with new IMEK. The steps are shown below:

Step 1 Communication initialization command

Use IMEK, with the key type is 0x01 and the key index is 0x00

=>Generate IMEK authentication session key

Step 2 Two-way authentication command

Use IMEK, with the key type is 0x01 and the key index is 0x00

=>Use IMEK authentication session key for authentication

Step 3 Key generation command

Replace IMEK, with the key type is 0x01 and the key index is 0x00

=>Use the previous IMEK authentication session key to transmit the new IMEK

=>The previous IMEK will be replaced

The above steps are also applicable to the replacement of IAEK, MEK and AEK

11.6 Card Reader Shielding

If the terminal discovers that the card reader has failed for some reason, then the terminal will send the card reader shielding command to the card reader. In this point, the card reader will clear buffer zone, MEK, AEK, Msession and Asession and prohibits card-seeking function.

Normally, the terminal shall not send the card reader shielding command to the card reader. However, if the terminal detects that the card reader is a forgery, the terminal will send the card reader shielding command to the card reader.

Table 21 Card reader shielding command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x23
Data Length	Length of request message	0x00 0x00
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 22 Card reader shielding response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4)
Command code	Command type identifier	0x23
Data Length	Length of response message	0x00 0x01

Field	Meanings	Value
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_ACCESS_NOT_PERFORMED • RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

12 Transaction Message

This chapter describes various transaction messages transmitted between the terminal and the card reader in detail.

Transaction message	
Quick debit/credit contactless transaction	0x30
Reset transaction	0x31
Display status	0x32
Online post-processing	0x6A

12.1 Quick Debit/Credit Contactless Transaction

The terminal adopts the quick debit/credit contactless transaction command to inform the cardholder of card reader to get ready for using the card reader to perform quick debit/credit payment transaction. At the moment, the card reader shall enable card-seeking function and be ready to initiate the contactless payment process. After the card reader completes collecting contactless card data, it shall send the data to the terminal within the timeout constraint.

Table 23 Quick debit/credit contactless transaction command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x30
Data Length	Length of request message	0x00 0x06
Data field	Transaction amount The data format shall meet the UICS requirements	BCD code of 612-byte value
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 24 Quick debit/credit contactless transaction response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x30
Data Length	Length of response message	Variable
Response code	The response code of card reader returns RC_DATA indicating that the data field contains effective data	One of the following values: • RC_DATA • RC_FAILURE • RC_INVALID_DATA • RC_DDA_AUTH_FAILURE • RC_NO_CARD • RC_AUTH_NOT_PERFORMED • RC_MORE_CARDS • RC_Other_AP_CARDS • RC_US_CARDS • RC_SECOND_APPLICATION (When the card reader synchronously supports transaction application and non-transaction application, such as "non-payment operation" card application, the card reader shall use this response code to inform the UICS terminal that the transaction application has been completed, and the card reader is waiting for the terminal to initialize non-transaction application.)

Field	Meanings	Value
Data field	If the response code returned by card reader is RC_DATA, the data field must include the date and time information and the magnetic track data in the following format: • Identification number of scheme - X(1) • Time and date - X(14) adopt YYYYMMDDHHMMSS (BC D format) • Existence of magnetic track 1 - X(1) -0xD1, if magnetic track 1 exists	
	– Length of magnetic track 1 - X(1) – Data of magnetic track 1 (ASCII, maximum 76 bytes) • Existence of magnetic track 2 - X(1) – 0xD2, if magnetic track 2 exists – Length of magnetic track 1 - X(1) – Data of magnetic track 2 (BCD, maximum 19 bytes) – The magnetic track data can be directly read from the peer data in magnetic track 2 of the chip, with the tag 57 • Existence of chip data - X(1) – 0xD3, if chip data exists – It is followed by data length and chip data • Existence of other data – 0xD4 – It is followed by data length and added data - X(n)	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

Notice:

1. The chip data includes the following data:
 - Authorized amount, tag 9F02
 - Other amount, tag 9F03
 - Terminal country code, tag 9F1A
 - Terminal verification result (TVR) ("00..00" in default), tag 95
 - Transaction currency code, tag 5F2A
 - Transaction date, tag 9A
 - Transaction type ("00" in default) Tag 9C
 - Random number, tag 9F37
 - PAN serial number, tag 5F34
 - Application interactive profile (AIP), tag 82
 - Application transaction counter (ATC), tag 9F36
 - Application cryptogram (AC), tag 9F26
 - Cryptogram information data, tag 9F27
 - Issuer's application data, tag 9F10
2. The "other data" field includes the following data:
 - a) Electronic cash issuer authorization code in TLV format, tag 9F74
 - b) Effective offline transaction amount in TLV format, tag 9F5D
 - c) Online transaction PIN identification, tag 99, followed by the online PIN data format using ISO 9564-1 format 0.

The ISO 9564-1 format 0 is shown in the table below:

Table 25 ISO9564-1 format

Format	Bit 1	Bit 5	Bit 9	Bit 13	Bit 17	Bit 21	Bit 25	Bit 29	Bit 33	Bit 37	Bit 41	Bit 45	Bit 49	Bit 53	Bit 57	Bit 61
0	0000	N	P	P	P	P	P/F	P/F	P/F	P/F	P/F	P/F	P/F	P/F	F	F

Wherein:

- N = PIN Length, 4 binary digits; its value range is 0100 ~ 1100
- P = PIN value, 4 binary digits; its value range is 0000 ~ 1001
- P/F = PIN/Filler; whether these Bits are PIN or padding Bits depends on the PIN length field.
- P/T = PIN/Transaction digit; whether these Bits are PIN or transaction value depends on the PIN length field.
- F = Filler, 4 binary digits; its value is 1111
- T = Transaction value, 4 binary digits; its value range is 0000 ~ 1111

For example, if the value of PIN is 1234, the data shall be 99 08 04 12 34 FF FF FF FF FF

If the card requires online PIN, but the card reader doesn't support online PIN, the card reader must return the tag 99 followed by 00, representing that no online PIN is supported. The card reader must transmit the data in TLV format (99 01 00).

The online PIN must be encrypted with different acquirer keys. These keys can be distributed in similar path, just like the distribution of Msession. In other words, the master keys as $IMEK_{MDK}$ shall be ready for derivation of keys.

d) Other CVM

- Offline plaintext PIN, tag 44; when the result is 01, it indicates the verification is successful; when the result is 00, it indicates that it isn't supported.
- DDA failure indication, tag 01; it must be forwarded when the DDA verification failed. When the result is 01, it indicates that DDA failed, the issuer is required to send the offline cryptogram (TC) for online authorization. When the DDA verification is successful, it shall not forward the tag.
- Signature, tag 55, the result is 01 00.

Notice: All data between the terminal and the card reader is transmitted in TLV format.

Notice: The third-party application or the application of other payment schemes can adopt other fields to transmit chip data.

12.2 Complete UICS Debit/Credit Transaction (Optional)

Table 26 Complete UICS debit/credit transaction request command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)

Field	Meanings	Value
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x3A
Data Length	Length of request message	Depending on the different phases, the data length varies. Refer to:
		Table 28 Complete UICS deb- it/credit transaction process
Data field	X(1) Process identification X(n) Terminal data	See Table 28 for the complete UICS process
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 27 Complete UICS debit/credit transaction response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x3A
Data Length	Length of response message	Variable
Response code	The response code of card reader returns RC_DATA indicating that the data field contains effective data	One of the following values: - RC_SUCCESSDATA - RC_FAILURE - RC_INVALID_DATA - RC_DDA_AUTH_FAILURE - RC_NO_CARD - RC_AUTH_NOT_PERFORMED - RC_MORE_CARDS
Data field	X(1) Process identification X(n) Card reader data	See Table 28 for the complete UICS process
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

Table 28 Complete UICS debit/credit transaction process

Process terminal data	Process identification (Hex)	Process Reader data
X(1) Mandatory online flag 00 - No mandatory online 01 - Mandatory online X (612) Transaction amount (BCD) The data format shall meet the UICS requirements	0x01: The terminal initiates transaction and transmits transaction. Amount	
	0x02: The card reader returns candidate application list	Notice: This process appears only when the terminal is required to perform manual selection with the existence of multiple candidate lists. Otherwise, the card reader directly returns the candidate application list of transaction request data 0x0A: X(1): Number of candidate applications, m X1(1): AID length, n1 X1(n1): AID X1(1): Length of application tag, n2 X1(n2): Application tag
		X1(1): Length of application preferred name, n3 X1(n3): Application preferred name X1(1): Identification of application priority identifier X1(1): Application priority identifier ... Xm(1): AID length, n1 Xm(n1): AID Xm(1): Length of application tag, n2 Xm(n2): Application tag Xm(1): Length of application preferred name, n3 Xm(n3): Application preferred name Xm(1): Identification of application priority identifier Xm(1): Application priority identifier
X(1) AID length, n X(n)AID	0x03: The terminal sends the selected	

Process terminal data	Process identification (Hex)	Process Reader data
Or X(1) 0xFF: The terminal cancels transaction	AID	
	0x0A: The card reader returns the transaction request type	X(1) Transaction result: 00 - Offline approval; 01 - Transaction rejection; 02 - Online request. If the transaction result is offline approval or online request, the data field shall return these data and encode them in TLV format: • Terminal Verification Result (TVR), tag 95 • Transaction date, tag 9A • Random number, tag 9F37 • PAN, tag 5A • PAN sequence number, tag 5F34 Application interchange profile (AIP), tag 82 • Application transaction counter (ATC), tag 9F36 • Cryptogram information data (CID), tag 9F27 • Application cryptogram (AC), tag 9F26 • Issuer's application data, tag 9F10 • Equivalent data in magnetic track 2, tag 57 • Cardholder verification method - Require online cryptogram PIN: 99 01 00 - Require signature: 55 01 00
– X(1): Online result: 0x00 - Normal online; 0x01 - can't get online – X (n): authorization response code, tag 8A, TLV format – X (n): authorization code, tag 89, TLV format	0x0B: The terminal transmits the online result	
– X (n): issuer authorization data, tag 91, TLV format – X (n): 71 script data, tag 71, TLV format – X (n): 72 script data, tag 72, TLV format		

Process terminal data	Process identification (Hex)	Process Reader data
	0x99: The card reader returns transaction completion.	X (1) Transaction result, 00 – transaction approved 01 - Transaction rejected X(1) Length of script execution result, the multiple of 5 X(n) Script execution result, see the debit/credit specification for the format When the transaction result is approval, these data shall be returned and encoded in TLV format: - Terminal Verification result (TVR), tag 95 - Application transaction counter (ATC), tag 9F36 - Cryptogram information data (CID), tag 9F27 - Application cryptogram (AC), tag 9F26 - Issuer's application data, tag 9F10

Note:

In the case that multiple candidate lists are present, the default value of timeout in which the card reader waits for the terminal to select application is 20 seconds.

In the case that the online request is required, the default value of timeout in which the card reader waits for the terminal to return online result is 20 seconds.

12.3 Application Identification

The scheme identification (scheme ID) is a value of one byte length, and it is used to indicate payment scheme and sub-scheme. The first half byte is used to indicate payment scheme, while the second half byte is used to indicate payment sub-scheme.

Payment application	Payment sub-application	HLB* value
UnionPay	UICS	0x90
UnionPay	QUICS	0x91
UnionPay	MSD	0x92
UnionPay	UPCARD	0x93

12.4 Reset

The reset command is used to clear the buffer zone of card reader. It can be executed in the following situations:

- The terminal sends transaction payment preparation command for successive 3 times with the same sequence number, but doesn't receive any response information from the card reader.
- Press the "cancellation" key. The terminal has sent the transaction payment preparation command to the card reader to inform the card reader to be ready for contactless payment transaction. However, the transaction is terminated due to some reasons. If the sequence number used in the transaction payment preparation command is [n], the sequence number [n+2] shall be used in the reset command.
- Return to normal mode from management mode or debugging/optimization mode.
- Stop the card reader from performing contactless payment transaction.

The card reader must terminate card-seeking function and wait for the next transaction payment preparation command.

Table 29 Reset command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x31
Data Length	Length of request message	0x00 0x01
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 30 Reset command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)

Field	Meanings	Value
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x31
Data Length	Length of response message	0x00 0x01
		One of the following values:
		RC_SUCCESS
Response code	Response code of card reader	RC_FAILURE
		RC_INVALID_COMM AND
		RC_AUTH_NOT_PERF ORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

12.5 Display Status

The terminal uses the display status command to request the card reader to display the status of certain operation and point out cardholder information.

Table 31 Display status command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x32
Data Length	Length of request message	0x00 0x01 Variable

Field	Meanings	Value
Data field	Status information – X(1): 0x00 - Success 0x01 - Failure – X(1): Total number of prompt information ID – X(1): Prompt information ID 1 – X(1): Prompt information ID 2 – X(1): Prompt information ID N	Variable data
	The maximum of N is 20	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 32 Display status command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x32
Data Length	Length of response message	0x00 0x01
		One of the following values:
		RC_SUCCESS
Response code	Response code of card reader	RC_FAILURE
		RC_INVALID_COMMAND
		RC_INVALID_DATA
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

Once the card reader receives this command, the actual action is subject to the own configuration of card reader, such as the number of LED, properties of buzzer and other factors.

12.6 UPCARD Transaction Processing

When the terminal sends the UPCARD transaction command, the card reader shall initiate the card-seeking function. If the UPCARD card is read, then return the obtained transaction data to the terminal for processing; if failed, then return.

Table 33 UPCARD transaction command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x6B
Data Length	Length of request message	0
Data field	None	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 34 UPCARD transaction command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x6B

Field	Meanings	Value
Data Length	Length of response message	Variable
Response code	Response code of card reader Return RC_SUCCESS, indicating that the data field contains effective data	One of the following values: • RC_SUCCESS • RC_FAILURE
Data field	If the response code returned by the card reader is RC_SUCCESS, the data field must return data in the following format: • Data length - X(2) • Response data - X(n)	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

13 High-level Transaction Process

This chapter describes the high-level transaction process between the terminal and the card reader.

Notice: The order of powering on terminal and card reader makes no difference. It is acceptable that either the terminal or the card reader is powered on first. After power-on, the terminal sends POLL command to the card reader, and the card reader makes a response. After data connection between the terminal and the card reader is established, the terminal starts the two-way authentication with communication initialization command and initiates a secure session. Then the two-way authentication command is used to initialize the authentication process.

Every time when the terminal and card reader are powered on, the transaction process between devices can be started only after the two-way authentication is successful.

13.1 Start-up Process

Terminal	Data flow direction	Card Reader
Power-on		Power-on
POLL	→	
	←	POLL_P (Respond within 0.5s)
Communication initialization command	→	
	←	Response (Respond within 0.5s)
Two-way authentication command	→	
	←	Response (Respond within 0.5s)

If the terminal and card reader are powered on for the first time, the terminal will be required to send two key generation commands for the generation of MEK and Msession keys, as shown below:

Key generation command (if the terminal and card reader establish data connection for the first time, it will be used to generate MEK)	→	
	←	Response (Respond within 0.5s)

For the follow-on power-on processing, only the generation of Msession is required:

Key generation command (generate Msession)	→	
		Response (Respond within 0.5s)

The terminal sends POLL command to the card reader in each P_POLL_MSG period to inquire whether the card reader is present:

POLL	→	
	←	POLL_A (Respond within 0.5s)
POLL	→	
	←	POLL_A
POLL	→	
	←	POLL_A

13.2 Successful Quick Debit/Credit Contactless Payment Process

Terminal	Data flow direction	Card Reader
POLL	→	
	←	POLL_A
POLL	→	
	←	POLL_A
Quick debit/credit contactless transaction command (recommended timeout of 30s)	→	
		Response: • If the card reader detects a card, then the data in the card will be rapidly obtained • When removing the card, the card reader must process the card data • The card reader shall immediately
	←	

		return response data after the card data is completely processed.
Display status	→	
	←	Response (0.5s)

After the transaction is completed, the terminal still transmit the POLL command to the card reader, and the card reader makes a response with POLL_A.

POLL	→	
	←	POLL_A

13.3 Successful and Complete UICS Debit/Credit Transaction Process (Optional)

See Table 28 - complete UICS debit/credit transaction process in Section 12.2 complete UICS debit/credit transaction.

13.4 Failed Transaction Process

The example of failed transaction process is given out as below:

If the terminal sends out the quick debit/credit contactless transaction command or complete UICS debit/credit transaction command, but the card reader doesn't detect any card, then the following process will be performed:

terminal	Data flow direction	Card Reader
POLL	→	
	←	POLL_A
POLL	→	
	←	POLL_A
Quick debit/credit contactless transaction command or complete UICS debit/credit transaction command	→	

If the terminal sends out the quick debit/credit contactless transaction command/complete UICS debit/credit transaction command, but the card reader doesn't detect any card within 5s, the response information with the response code RC_NO_CARD shall be returned to the terminal. Then, the terminal shall send the same command for 2 times. If the card reader doesn't make any response or the response code is RC_NO_CARD, the terminal shall send the display status command to the card reader and the card reader shall display information for timeout prompt. The merchant shall be able to use the cancel key to send transaction reset command to the card reader, in order to complete the transaction at any time. At the moment, the card reader shall prohibit card-seeking function.

After that, the terminal still sends POLL command to the card reader, and the card reader responds to it with POLL_A.

POLL	→	
	←	POLL_A

13.5 No Connection of Card Reader Connecting Wire

13.5.1 Pull-out and Reconnection of Card Reader Connecting Wire

Case I: If the connecting wire is pulled out and then reconnected within the time interval between POLL_A and next POLL, it won't be necessary to confirm data connection again. The flow is as follows:

terminal	Data flow direction	Card Reader
POLL	→	
	←	POLL_A

Case II: The connecting wire is pulled out and isn't reconnected within the time interval between POLL_A and next POLL.

terminal	Data flow direction	Card Reader
POLL	→	
POLL	→	
POLL	→	

After the terminal sends POLL for 3 times, if the card reader still makes no response, the terminal will send POLL command at the interval of P_POLL_MSGs period.

POLL	→	
	←	POLL_P

If the POLL_P response is received, the terminal will confirm that the card reader is connected, and it will be necessary to establish new data connection according to the steps in Section 13.1.

13.5.2 Sudden Power Failure then Power-on of Card Reader

terminal	Data flow direction	Card Reader
POLL	→	
POLL	→	
POLL	→	

After the terminal sends POLL for 3 times, if the card reader still makes no response, the terminal will send POLL command at the interval of P_POLL_MSGs period.

POLL	→	
	←	POLL_P

After that, the card reader is powered on and the terminal shall perform the polling. The terminal is required to establish a new data connection as per the steps in Section 13.1.

(If the power of card reader fails and is powered on again within the time interval between POLL_A and next POLL, the terminal can't detect this process. However, if the terminal sends POLL command and the card reader shall respond POLL_P, then the terminal can judge that the card reader hasn't performed two-way authentication. At the moment, it is required to establish a new data connection as per the steps in Section 13.1.)

14 Management Message

14.1 Entering Management Mode

The entering management mode command is the precondition for implementation of other management commands. Before applying this command, please confirm that the access condition has been met. The response data shall contain the following contents:

- Manufacturer logo
- Firmware Version No.
- Reserved data

This command can allow the card reader to enter management mode or to exit management mode and enter normal mode.

Table 35 Entering management mode command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x40
Data Length	Length of request message	0x00 0x01
Data field	X(1) • 0x00 Set management mode • 0x01 Set normal mode	X(1)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 36 Entering management mode command response

Field	Meanings	Value
STX	Start of message	0x02

Field	Meanings	Value
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x40
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: ● RC_SUCCESS ● RC_FAILURE ● RC_INVALID_COMMAND ● RC_ACCESS_FAILURE
Data field	● X(8) - Manufacturer identification ● X(4) - Firmware version number ● X(4) - Reserved data	X(10)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.2 Obtain Properties

This command is used to obtain the information of payment scheme and sub-scheme in the card reader.

Table 37 Obtain properties command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x41

Field	Meanings	Value
Data Length	Length of request message	Variable
Data field	X(1) - "FF": Request the card reader to show the list of all supported payment schemes.	Variable
	If X(1) isn't "FF", the value of X(1) is the number of payment schemes.	
	X(n) - Scheme ID (The number of X(n) depends on the above number of schemes.)	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 38 Obtain properties command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x41
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: • RC_SCHEME_SUPPORTED • RC_INVALID_SCHEME • RC_INVALID_DATA • RC_ACCESS_NOT_PERFORMED

Field	Meanings	Value
Data field	If the response code is RC_SCHEME_SUPPORTED, the data field shall contain the following data: • X(1) - Number of schemes • X(1) - Scheme ID • X(1) - 00 Not supported – 01: Supported • X(1) - Scheme ID • X(1) - 00 Not supported – 01: Supported ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.3 Set Properties

The terminal uses this command to activate or prohibit one or more payment schemes/payment sub-schemes supported by the card reader.

The card reader described in the Specification shall support the following payment schemes as a default:

Payment scheme	Payment sub-scheme	HLB* value
CUP	UICS	0x90
CUP	QUICS	0x91
CUP	MSD	0x92
CUP	UPCARD	0x93

Table 39 Set properties command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01

Field	Meanings	Value
Command code	Command type identifier	0x42
Data Length	Length of request message	Variable
Data field	• X(1) - Number of schemes • X(1) - Scheme ID • X(1) - 00 Prohibit this payment scheme/sub-scheme – 01 Activate this payment scheme/sub-scheme • X(1) - Scheme ID • X(1) - 00 Prohibit this payment scheme/sub-scheme – 01 Activate this payment scheme/sub-scheme ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 40 Set properties command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x41
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_INVALID_SCHEME • RC_ACCESS_NOT_PERFORMED

Field	Meanings	Value
Data field	• X(1) - Number of schemes • X(1) - Scheme ID • X(1) - 00 Prohibit this payment scheme/sub-scheme – 01 Activate this payment scheme/sub-scheme • X(1) - Scheme ID • X(1) - 00 Prohibit this payment scheme/sub-scheme – 01 Activate this payment scheme/sub-scheme ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.4 Obtain Time and Date

This command is used to obtain the time and date information in the card reader.

Table 41 Obtain time and date command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x43
Data Length	Length of request message	0x00 0x00
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 42 Obtain time and date command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x43
Data Length	Length of response message	0x00 0x08
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMM AND • RC_ACCESS_NOT_P ERFORMED
Data field	Format of time and date: YYYY MM DD HH MM SS	N(14)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC
EXT	End of message	0x03

14.5 Set Time and Date

This command is used to set the time and date information in the card reader.

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x44
Data Length	Length of request message	0x00 0x07

Field	Meanings	Value
Data field	Format of time and date: YYYY MM DD HH MM SS	N(14)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 44 Set time and date command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x44
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMM AND • RC_ACCESS_NOT_PERFORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.6 Obtain Parameters

This command is used to obtain various different preloaded parameters in the card reader.

The preloaded parameters are described in Appendix A in detail.

Table 45 Obtain parameters command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x45
Data Length	Length of request message	0x00 0x02
Data field	Parameter Index	X(2)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 46 Obtain parameters command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x45
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_DATA • RC_INVALID_PARAM • RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, the parameter information contained in the data field shall follow the below format: X(2) - Parameter index X(2) - Length of parameter data	Variable

Field	Meanings	Value
	X(n) - Parameter data	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.7 Obtain Serial Interface Communication Rate

This command is used to obtain the serial interface communication rate used by the card reader.

Table 47 Obtain serial interface communication rate command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x52
Data Length	Length of request message	0x00 0x00
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 48 Obtain serial interface communication rate command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x52
Data Length	Length of response message	0x00 0x01 or 0x00 0x02

Field	Meanings	Value
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_INVALID_DATA • RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the returned data information is shown as below: • X(1) - – 00: 115200 baud – 01: 57600 baud – 02: 38400 baud – 03: 28800 baud – 04: 19200 baud – 05 ~ 10 RFU	X(1)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.8 Set Serial Interface Communication Rate

This command is used to set the serial interface communication rate for the card reader.

Table 49 Set serial interface communication rate command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x53
Data Length	Length of request message	0x00 0x01

Field	Meanings	Value
Data field	The serial interface communication rate shall be set in the following format: • X(1) - – 00: 115200 baud – 01: 57600 baud – 02: 38400 baud – 03: 28800 baud – 04: 19200 baud – 05 ~ 10 RFU	X(1)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 50 Set serial interface communication rate command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x53
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_INVALID_DATA • RC_ACCESS_NOT_PERFORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.9 Reset Acquirer Key

This command is used to clear MEK and AEK.

It shall be noted that the security access condition must be satisfied when this command is used.

Table 51 Reset acquirer key command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x54
Data Length	Length of request message	0x00 0x02
Data field	X(1) - Key type • “0x02” MEK • “0x06” AEK • X (1) - Key index	X(2)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 52 Reset acquirer key command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x54
Data Length	Length of response message	0x00 0x01

Field	Meanings	Value
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_ACCESS_NOT_PERFORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.10 Resume Card Reader

This command is used to enable the card-seeking function of card reader.

It shall be noted that the security access condition must be satisfied when this command is used.

Table 53 Card reader resume command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x55
Data Length	Length of request message	0x00 0x00
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 54 Card reader recovering resume response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x55
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_ACCESS_NOT_PERFORMED
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.11 Obtain UICS Tag Value

This command is used to obtain the UICS data element tags supported by the card reader as well as their data.

Table 55 Obtain UICS tag value command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x56
Data Length	Length of request message	Variable
Data field	X(1) - "FF" Request for all tags and data X(n) - Tag	Variable

Field	Meanings	Value
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 56 Obtain UICS tag value command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x56
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_INVALID_DATA • RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the format of contained data is shown as follow: • X(1) - "FF" (It indicates the request for all tags and data) or X(1) - No "number of tags". • X (n) - TLV format (See the UICS Specification for the details.) (When the tag doesn't contain data, V doesn't exist and L = "00".)	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.12 Set UICS Tag Value

This command is used to set the value of UICS data element tag of the card reader. It is used only when the terminal and the card reader are connected for the first time.

Table 57 Set UICS tag value command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x57
Data Length	Length of request message	Variable
Data field	• X(1) - Number of tags • TLV format (See the UICS Specification for the details.)	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 58 Reset acquirer key command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x57
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_NO_UICS_TAGS

Field	Meanings	Value
		- RC_INVALID_DATA - RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the following data shall be contained: - X(1) - Number of tags - TLV format (See the UICS Specification for the details.)	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.13 Obtain Display Information

This command is used to obtain the prompt information stored by the card reader. These information is shown in the table below:

Table 59 Prompt information

Information ID	Information definition	Examples
1	Welcome information	Welcome
2	Fixed thanks information	THANK YOU
3	Dynamic thanks information (The acquirer can customize this information in a dynamic way. For example, the name of cardholder can be shown.)	Thank you, tag 5F20 (The tag 5F20 represents the name of cardholder). For example: Thank you, Wang Liang.
4	Transaction has been completed in the online or offline mode.	Transaction successful
5	DDA authentication failed. Transaction is rejected by the issuer or transaction is rejected due to other unknown reasons.	Please use other cards.
6	Exceed the contactless transaction limit	Please insert card
7	Several cards are present in the sensing area	Please use one card
8	The cardholder moves the card so fast that the card reader did not obtain all the required transaction data	Please retry
9	RFU	RFU

Information ID	Information definition	Examples
10	Require a signature on the transaction receipt printed by the terminal	Please sign on the purchase slip
11	Require inputting PIN	Please input the password
12	Offline transaction limit	Offline transaction limit
13	Require inputting PIN. Transaction has not been completed	Transaction has not been completed. Please input PIN
14	Require a signature. Transaction has not been completed	Transaction has not been completed. Please sign your name
15	The card reader isn't ready	The card reader isn't ready
16	Prompt the cardholder to show the card	Please swipe card
17	Card data has been read. Wait for completion of authentication/authorization	Please wait for a moment
18	If the card reader executes only the payment transaction, the cardholder can be informed of this information	processing
19	RFU	-
20	RFU	-
21	RFU	-
22	RFU	-
23	RFU	-
24	RFU	-
25	RFU	-
26	RFU	-
27	RFU	-
28	RFU	-

Table 60 Obtain display information command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)

Field	Meanings	Value
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x58
Data Length	Length of request message	Variable
Data field	- X(1) - "FF" Request for all prompt information - X(1) - Prompt information ID	X(1)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 61 Obtain display information command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x58
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE - RC_INVALID_COMMAND - RC_INVALID_DATA - RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the following data shall be contained: - X(1) - No total number of prompt information - X(n) - Prompt information ID + Prompt information Length + contents of prompt information	Variable
CRC	See Section 7.5.7 for the calculation	Calculated CRC value

Field	Meanings	Value
	method	
EXT	End of message	0x03

14.14 Set Display Information

The terminal sends display information to the card reader by this command. This command is applicable to the card reader which only supports the character and numerical information in ASCII format. If the card reader requires displaying the specialized information formats of other languages, the card reader manufacturer must provide independent tool to update these information.

Table 62 Set display information command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x59
Data Length	Length of request message	Variable
Data field	- X(1) - Total number of prompt information - X(1) - Prompt information ID (1-30) - X(1) - Length of prompt information. - X(n) - Contents of prompt information ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 63 Set display information command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x59
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_INVALID_DATA - RC_NO_UICS_TAGS - RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the following data shall be contained: - X(1) - Total number of prompt information - X(n) - Prompt information ID + Prompt information Length + contents of prompt information	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.15 Obtain CVM Properties

This command is used to obtain the CVMs (cardholder verification methods) supported by the card reader.

Table 64 CMV list

CVM	CVM ID	Status
Not support CVM	0x00	0x00
Signature	0x10	0x00 : Activating 0x01: Releasing
Online PIN	0x11	0x00 : Activating 0x01: Releasing

Table 65 Obtain CVM properties command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x5A
Data Length	Length of request message	0x01
Data field	X(1) - "FF" Request for all supported CVM lists	0x01
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 66 Obtain CVM properties command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x5A
Data Length	Length of response message	Variable
Response code	Response code of card reader	One of the following values:
		- RC_SUCCESS - RC_FAILURE - RC_INVALID_COMMAND - RC_ACCESS_NOT_PERFORMED
Data field	If the response code is RC_SUCCESS, then the following data shall be contained: - X(1) - Number of supported CVM - X(2) CVM ID and status	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value

Field	Meanings	Value
EXT	End of message	0x03

14.16 Set CVM Properties

This command is used to activate or prohibit CVM properties. All CVM properties in the card reader are prohibited as a default. They can be activated only when the terminal sends this command.

Table 67 Set CVM properties command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x5B
Data Length	Length of request message	Variable
Data field	- X(1) - Number of CVM - X(1) - CVM ID - 0x00 - Prohibiting – 0x01 - Activating - X(1) - CVM ID – 0x00 - Prohibiting – 0x01 - Activating ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 68 CVM Set properties command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x5B
Data Length	Length of response message	Variable

Field	Meanings	Value
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_INVALID_SCHEME - RC_ACCESS_NOT_PERFORMED
Data field	- X(1) - Number of CVM - X(1) - CVM ID – 0x00 - Prohibiting – 0x01 - Activating - X(1) - CVM ID – 0x00 - Prohibiting – 0x01 - Activating ...	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.17 Set Card Reader Public Key

This command is used to add or delete card reader public key. Only one public key can be added every time, or one or all public keys can be deleted every time. The added public key will overlay the original public key if this public key is included in the card reader.

Table 69 Set card reader public key command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x61
Data Length	Length of request message	Variable
Data field	- X(1) - 0x11 Add public key. 0x21 Delete public key as per RID and public key index 0x22 Delete public key as per RID.	Action type 1. Adding public key (0x11) Required data: RID+INDEX+DATALEN+DATA 2. Deleting single public key

Field	Meanings	Value
	- X (5) - card organization ID number - X (1) - Key index - X(2) - Length of key (for addition of key) - X(n) - Key data (for addition of key)	(0x21) Required data: RID+INDEX 3. Deleting public key as per RID (0x22) Required data: RID
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 70 Set card reader public key command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x61
Data Length	Length of request message	0x00 0x01
Response code	Response code of card reader	One of the following values:
		- RC_SUCCESS - RC_FAILURE - RC_INVALID_COMMAND - RC_INVALID_CA_KEY - RC_INVALID_DATA - RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

CA public key data follows the below format:

- X (2) - Public key data length
- X (1) - Hash algorithm identifier
- X (1) - Public key algorithm identifier

- X (1) - Public key modulus length
- X (n) - Public key module
- X (1) - Public key index length
- X (n) - Public key index
- X (20) – Hash value

14.18 Generic Search for Card Reader Public Key

This command is used to search for the public key stored in the card reader.

Table 71 Generic search for card reader public key command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x62
Data Length	Length of request message	0x00 0x06
Data field	• X (5) - card organization ID number • X (1) - Key index	X(6)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 72 Generic search for card reader public key command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x62
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS

Field	Meanings	Value
		RC_FAILURE
Data field	CA public key data	If the response code is RC_SUCCESS
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

CA public key data follows the below format:

- X (2) - Public key data length
- X (1) - Hash algorithm identifier
- X (1) - Public key algorithm identifier
- X (1) - Public key modulus length
- X (n) - Public key module
- X (1) - Public key index length
- X (n) - Public key index
- X (20) – Hash value

14.19 Set Card Reader Revocation Public Key certificate

This command is used to add or delete card reader revocation public key certificate. Only one revocation public key certificate can be added every time and one revocation public key certificate or all revocation public key certificates can be deleted every time. The added public key will overlay the original revocation public key certificate if this revocation public key certificate is included in the card reader.

Table 73 Set card reader revocation public key certificate command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x63

Field	Meanings	Value
Data Length	Length of request message	Variable
Data field	- X(1) - 0x11 Add revocation certificate 0x21 Delete revocation certificate 0x22 delete all The command for deleting all revocation certificates has no follow-on data field. - X (5) - card organization ID number - X (1) - Key index - X(3) - Sequence number	Variable
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 74 Set card reader revocation public key certificate command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x63
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE - RC_INVALID_COMMAND - RC_INVALID_DATA - RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.20 Search for Card Reader Revocation Public Key certificate

This command is used to search for the revocation public key certificate stored in the card reader.

Table 75 Search for card reader revocation public key certificate command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x64
Data Length	Length of request message	0x00 0x09
Data field	- X (5) - card organization ID number - X (1) - Key index - X(3) - Sequence number	X(9)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 76 Search for card reader revocation public key certificate command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x64
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.21 Set Card Reader Blacklist

This command is used to add or delete the blacklist stored in the card reader. Only one blacklist can be added every time, and one blacklist or all blacklists can be deleted every time. The added public key will overlay the original blacklist if this blacklist is included in the card reader.

Table 77 Set card reader blacklist command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x65
Data Length	Length of request message	Variable
Data field	- X(1) - 0x11 Add blacklist. 0x21 Delete blacklist. 0x22 delete all The command for deleting all blacklists has no follow-on data field. - X(1) - Number of blacklists - X(10) - Compressed card number - X(1) - Sequence number of card	- If it is the blacklist adding command, the card number and the sequence number of card will occur for several times depending on the number of blacklists - If it is the blacklist deleting command, the card number and the sequence number of card can occur for only one time - If it is the command for deleting all blacklists, no follow-on request data shall be present
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 78 Set card reader blacklist command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x65

Field	Meanings	Value
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: • RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_INVALID_DATA • RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.22 Search Card Reader Blacklist

This command is used to search the blacklist stored in the card reader.

Table 79 Search card reader blacklist command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x66
Data Length	Length of request message	0x00 0x0B
Data field	• X(10) - Compressed card number • X(1) - Sequence number of card	X(11)
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 80 Search card reader blacklist command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)

Field	Meanings	Value
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x66
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.23 Set UICS fixed parameters

This command is used to set the fixed parameters for UICS transaction of card reader. These parameters can also be set by applying the set UICS tag function.

Table 81 Set UICS fixed parameters command

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x67
Data Length	Length of request message	Variable
Data field	- X(2) - Merchant classification code (UICS tag 9F15) - X(6) - Acquirer identification (UICS tag 9F01) - X(2) - Terminal country code (UICS tag 9F1A) - X(2) - Terminal transaction currency code (UICS tag 5F2A) - X(1) - Terminal transaction currency index (UICS tag 5F36)	

Field	Meanings	Value
	- X(1) - RFU length - X(n) - RFU data	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 82 Set UICS fixed parameters command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x67
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values: - RC_SUCCESS - RC_FAILURE - RC_INVALID_COMMAND - RC_INVALID_DATA - RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

14.24 Set Card Reader AID Parameters

This command is used to set the AID supported by the card reader as well as the parameters corresponding to these AIDs. One AID parameter can be set every time. If there are several supported AIDs, it will be necessary to perform the set operation several times.

Table 83 Set card reader AID parameters command

Field	Meanings	Value
STX	Start of message	0x02

Field	Meanings	Value
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0B\x01\x0E\x01
Command code	Command type identifier	0x68
Data Length	Length of request message	Variable
Data field	X(1) - Action type: 0x11 add AID. 0x21 delete a single AID 0x22 Delete the AIDs with the same application type.	Notice: Adding AID Required data: All data. If no data item is present, perform padding with \x0.
	- X(1) - Application identification (Refer to Section 12.3 Application Identification for the definition.) - X(1) - AID length (HEX) - X(n) - AID data - X (1) - Partial selection flag (Does not support, Support) - X(4) - Terminal transaction qualifier (UICS tag 9F66) - X(11) - Contactless transaction limit - X(11) - Contactless offline limit - X(11) - Contactless CVM limit - X (1) - Terminal default DDOL length (HEX) - X (n) - Terminal default DDOL - X(1) - Whether any follow-on parameter exists 0x00 - Not exist 0x11 - UICS parameter exists 0x12 - RFU data exists 0x13 - UICS data and RFU data exist - X (1) - Terminal type (UICS tag 9F35) - X (3) - Terminal properties (UICS tag 9F33) - X (5) - Terminal additional properties (UICS tag 9F40) - X (2) - Terminal application version number (UICS tag 9F09) - X (11) - Randomly selected threshold	. - Deleting single AID: Required data: Application identification (1) + AID length (1) + AID data (n) Data (n) - Deleting all AIDs with application identification A Required data: Application identification (1): A - When the "X(1) - Whether any follow-on parameter exists" is 0, this AID parameter is the transaction parameter which supports QUICS only, and the absence of follow-on data is acceptable. If it is the AID which supports complete contactless UICS process, the follow-on UICS parameter shall exist. - If the manufacturer requires adding other data, the self-definition can be performed in RFU.

Field	Meanings	Value
	• X (3) - Randomly selected target percentage • X (3) - Randomly selected maximum target percentage • X (5) - Terminal action code TAC - Rejection • X (5) - Terminal action code TAC - Online • X (5) - Terminal action code TAC - Default • X(1) - RFU length, n • X(n) - RFU data	
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Table 84 Set card reader AID parameters command response

Field	Meanings	Value
STX	Start of message	0x02
Series number	Serial No.	X(1)
Reserved for use	Reserved for manufacturer use	X(4) Default value is \x0E\x01\x0B\x01
Command code	Command type identifier	0x68
Data Length	Length of response message	0x00 0x01
Response code	Response code of card reader	One of the following values:
		• RC_SUCCESS • RC_FAILURE • RC_INVALID_COMMAND • RC_INVALID_DATA • RC_ACCESS_FAILURE
Data field	None	None
CRC	See Section 7.5.7 for the calculation method	Calculated CRC value
EXT	End of message	0x03

Appendix A

(Normative Appendix)

Internal parameters of card reader

The following parameters must be optimized and set before the terminal and card readers are sold.

Value	Index	Length	Contents
P_MSG_TIMEOUT	0x0001	X(2)	Millisecond level timeout. The card reader must make a response to the message (except quick debit/credit contactless transaction message) sent by the terminal within the default time. The default value is 500ms or 0x01F4
P_SALE_TIMEOUT	0x0002	X(2)	The millisecond level timeout is the timeout for waiting after the terminal sends quick debit/credit contactless transaction message. The default value is 15000ms or 0x3A98 It shall be noted that the data processing time of card reader isn't allowed to exceed 8000ms. This 8000ms timeout is used for processing some special situations. For example, the cardholder can't find its card, or the card is dropped, or the card isn't placed in a correct position etc. The card reader shall obtain the data in the card within 1s (within 0.5s as recommended). After the card is removed, the card reader must verify data within 1.5s and send data to the terminal as soon as possible.
P_POLL_MSG	0x0003	X(2)	It is the millisecond level timeout in which the terminal sends next POLL message to detect the presence of card reader. The default value is 30s or 0x001E
P_BUF_TIMEOUT	0x0004	X(2)	If the terminal doesn't make a response to the response message for quick debit/credit contactless transaction message of card reader, This parameter shall indicate the waiting time for the card reader to clear buffer zone. The default value is 5000ms or 0x1388
P_ENCRYPTION	0x0005	X(1)	The following values are supported: 0x00 TDEA has been prohibited. The data is transmitted between the terminal and the card reader in the form of plaintext. It is only used for testing and debugging. 0x01 TDEA has been enabled

Value	Index	Length	Contents
P_DISPLAY	0x0006	X(1)	The following values are supported: 0x00 The terminal shall display all response codes of card readers. It is used for testing and debugging only 0x01 The terminal shall convert the response code of card reader into prompt information and display it.
P_MAX_BUF_SIZE	0x0007	X(2)	It is the maximum buffer zone allocated for request message and command message by the card reader. The default value is 1024 bytes or 0x0400
P_DOUBLE_DIP	0x0008	X(2)	This value is the interval at which the same card is allowed by the card reader to perform two successive transactions. This parameter is used to avoid that the card is connected to the card reader twice and the same transaction is executed twice without the awareness of cardholder. The default value is 5000ms or 0x1388
P_READER_INDEX	0x0009	X(2)	Card reader index
P_LANGUAGE	0x000A	X(n)	This parameter indicates the language type supported by the card reader and the language type of prompt information. The definition of language type hereon conforms to ISO639. The format of parameter is shown as follow: Tag 5F2D + Length + Language type + Status (01: Activated; 00: Prohibited) For example: If the card reader supports English and Chinese synchronously in compliance with ISO639: English = en = 65 6E and Chinese = zh = 7A 68. 1. The following data will be shown when only Chinese information is displayed: 5F 2D 06 65 6E 00 7A 68 01. The following data will be shown when Chinese information is displayed in the first line and English information is displayed in the second line: 5F 2D 06 7A 68 01 65 6E 01.
P_DISPLAY_S_MSG	0x000B	X(2)	It is the millisecond level timeout for displaying short prompt information. The default value is 2000ms or 0x07D0

Value	Index	Length	Contents
P_DISPLAY_L_MSG	0x000C	X(2)	It is the millisecond level timeout for displaying long prompt information. The default value is 5000ms or 0x1388
P_DISPLAY_SS_MSG	0x000D	X(2)	It is the millisecond level timeout for signing on touch screen. The default value is 10000ms or 0x2710.
P_DISPLAY_SR_MSG	0x000E	X(2)	It is the millisecond level timeout for signing on receipt. The default value is 5000ms or 0x1388.
P_DISPLAY_PIN_MSG	0x000F	X(2)	It is the millisecond level waiting timeout for PIN input. The default value is 10000ms or 0x2710.
P_DISPLAY_E_MSG	0x0010	X(2)	It is the millisecond level timeout for displaying error information. The default value is 3000ms or 0x0BB8.
Reserved for future	0x0012 ~ 0x0FFF	X(2)	Reserved for the use in future
Specialized parameter	0x1000 ~ 0xFFFF	X(2)	Special parameter for specialized scheme

Appendix B

(Normative Appendix)

Response code

The response codes and error codes returned by the card reader are shown as follows.

Response code	HLB* value	Usage	Terminal display
RC_SUCCESS	0x00	It is the Generic response code which indicates that the card reader successfully executes request command.	SUCCESS
RC_DATA	0x01	The data obtained from contactless card by the card reader is valid. It is used to initiate transaction.	CARD DATA
RC_POLL_A	0x02	The card reader confirms response. Two-way authentication is achieved between the terminal and the card reader.	POLL A
RC_POLL_P	0x03	The card reader confirms response. No two-way authentication is achieved between the terminal and the card reader.	POLL P
RC_SCHEME_SUPPORTED	0x04	The card reader supports this payment scheme.	SCHEME
RC_SIGNATURE	0x05	Require signature	SIGNATURE
RC_ONLINE_PIN	0x06	Require online PIN	PIN_ONLINE
RC_OFFLINE_PIN	0x07	Require offline PIN	PIN_OFFLINE
RC_SECOND_APPLICATION	0x08	Prompt that the card reader supports the second application (non-payment application)	SECOND_APP

Response code	HLB* value	Usage	Terminal display
RC_FAILURE	0xFF	Common error, error exists in the request message	FAILURE
RC_ACCESS_NOT_PERFORMED	0xFE	The access control for opening management mode has not been executed	NO ACCESS
RC_ACCESS_FAILURE	0xFD	The access control for opening management mode error	FAIL ACCESS
RC_AUTH_FAILURE	0xFC	Two-way authentication failed	FAIL AUTH
RC_AUTH_NOT_PERFORMED	0xFB	No two-way authentication	NO AUTH
RC_DDA_AUTH_FAILURE	0xFA	DDA authentication failed	CARD FAIL
RC_INVALID_COMMAND	0xF9	Command code error	NO MSG ID
RC_INVALID_DATA	0xF8	The data field of request message error	DATA INCORRECT
RC_INVALID_PARAM	0xF7	No parameter exists here	NO PARA
RC_INVALID_KEYINDEX	0xF6	When Asession or AEK has not been generated, the terminal requests the card reader to generate Msession key.	BAD KEYID
RC_INVALID_SCHEME	0xF5	The card reader doesn't support the scheme activated by the terminal.	NO SCHEME
RC_MORE_CARDS	0xF3	Multiple cards	MORE THAN 1 CARD
RC_NO_CARD	0xF2	No contactless card is presented	NO CARD
RC_NO_UICS_TAGS	0xF1	The card reader doesn't support this tag	NO TAGS
RC_NO_PARAMETER	0xF0	No parameter exists here	NO PARA

Response code	HLB* value	Usage	Terminal display
RC_POLL_N	0xEF	The card reader confirms the response. The card reader isn't ready.	POLL N
RC_NO_PIN	0xEC	PIN code not entered	No PIN
RC_NO_SIG	0xEB	The touch screen didn't obtain the signature	No Signature